Sequencing methods

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Volos Summer School 2026

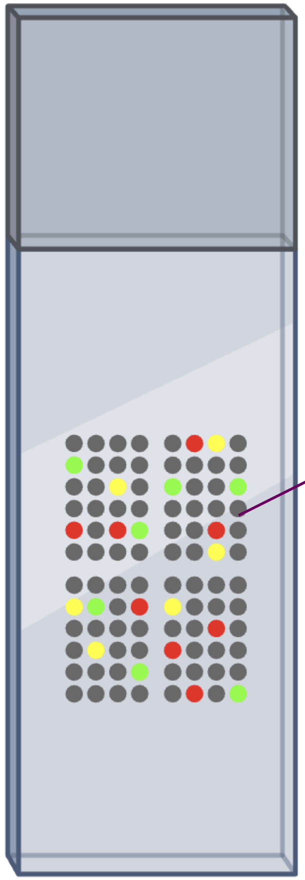
Sequencing technologies:

1. Genotyping arrays
2. NGS sequencing
3. Third-generation sequencing
4. RNA-sequencing
5. Single-cell RNA sequencing

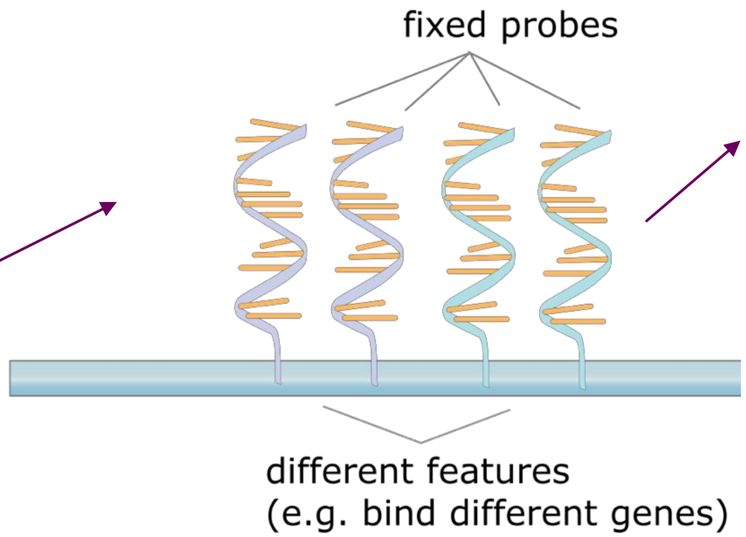
1. Genotyping arrays

- *Examine the individual's **DNA** sequence*
- *The **most widely used** genetic profiling technology in human genetics*

Genotyping arrays – what are they?



DNA microarray



Each probe interrogates ONE specific position - i.e. single nucleotide polymorphism **(SNP)**

Glass/silicon chip with hundreds of thousands to millions of short DNA sequences (***oligonucleotide probes***)

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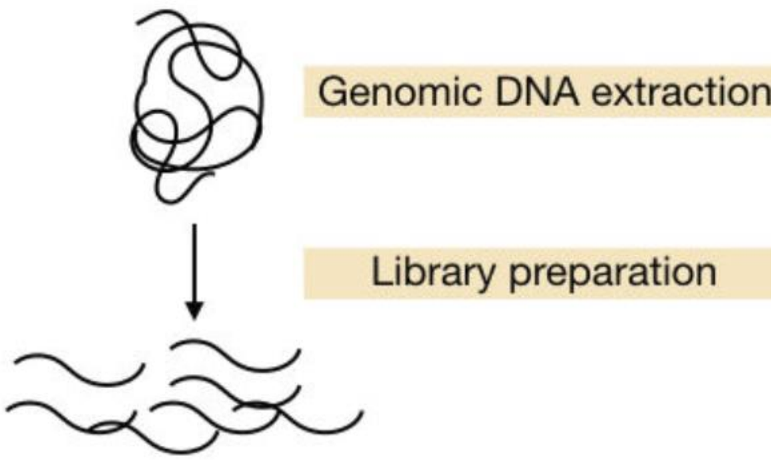
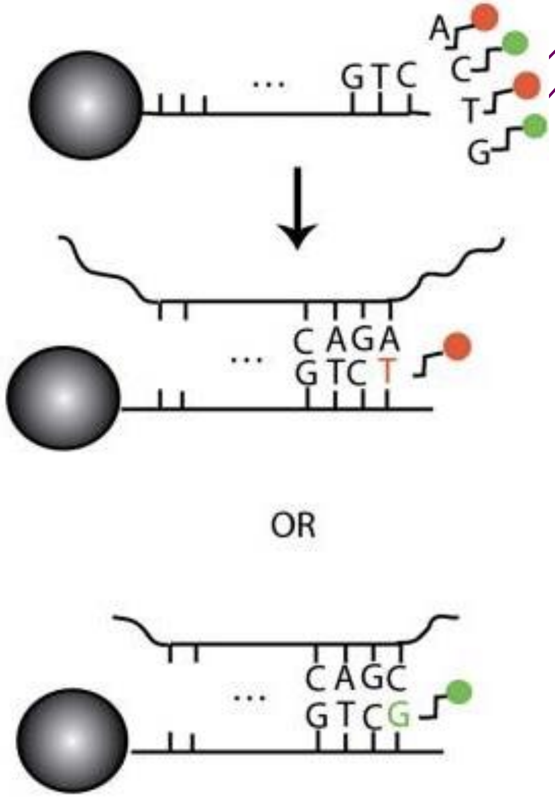
https://en.wikipedia.org/wiki/DNA_microarray#/media/File:NA_hybrid.svg

Genotyping arrays – how do they work?

Sample preparation, fragmentation and labeling



fragment sample DNA



Hybridization

fluorescent dyes

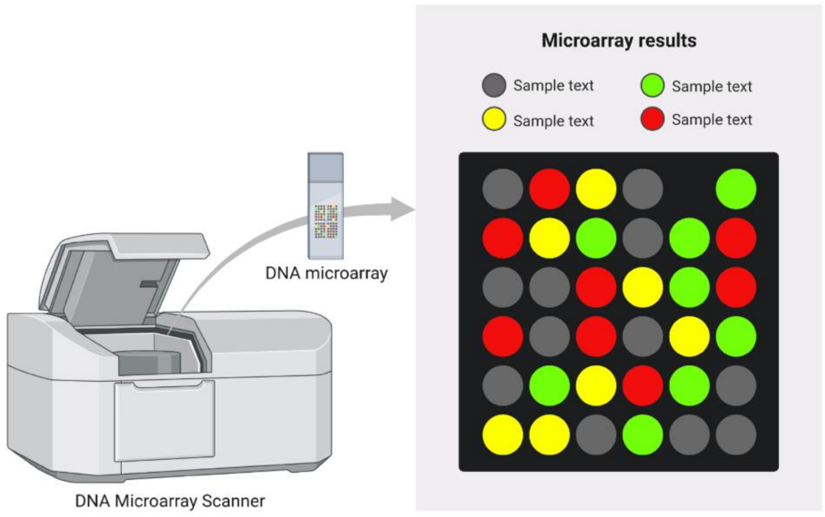
DNA **matches a probe**

↓

gets inserted in the sequence (hybridizes)

↓

fluoresces



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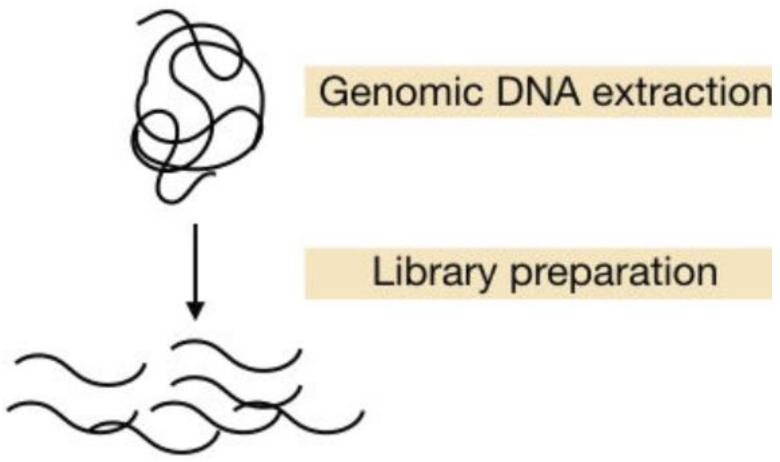
Genotyping arrays – how do they work?

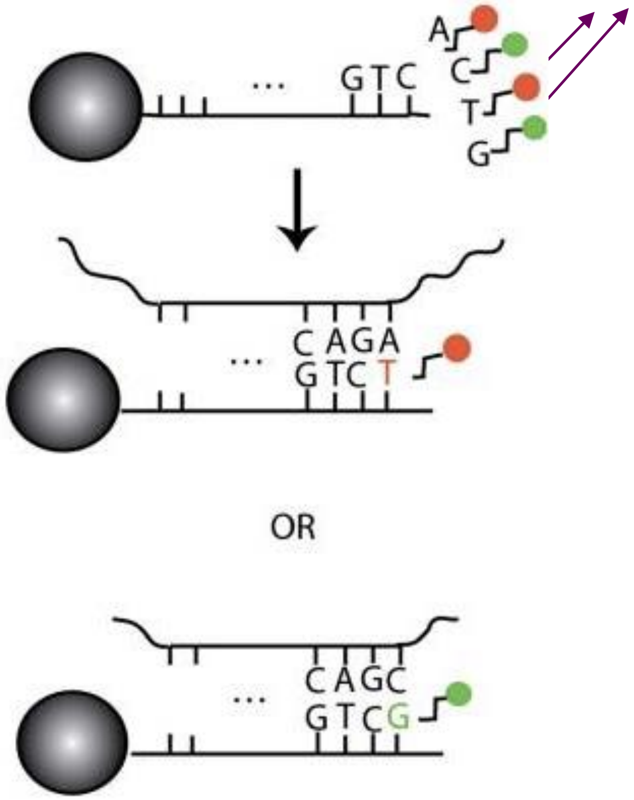
Sample preparation, fragmentation and labeling



fragment sample DNA

Hybridization

fluorescent dyes



Limitations:

- SNP frequencies are very different from one population to another
  - Every significantly different population will need a custom array
- Rare variants
  - Difficult to genotype
  - Poorly imputed
- De novo mutations
- Chromosome rearrangements not detectable (e.g. inversions)

Arrays can capture only **KNOWN** genetic variation

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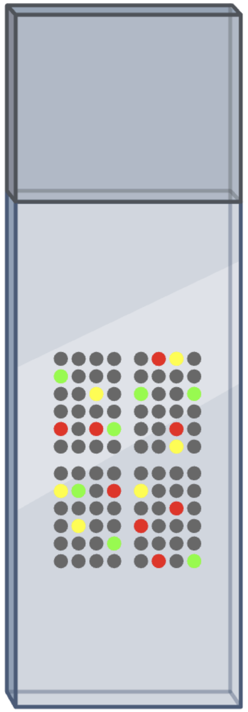
https://en.wikipedia.org/wiki/DNA_microarray#/media/File:NA_hybrid.svg

Imputation and Linkage Disequilibrium (LD)



DNA microarray

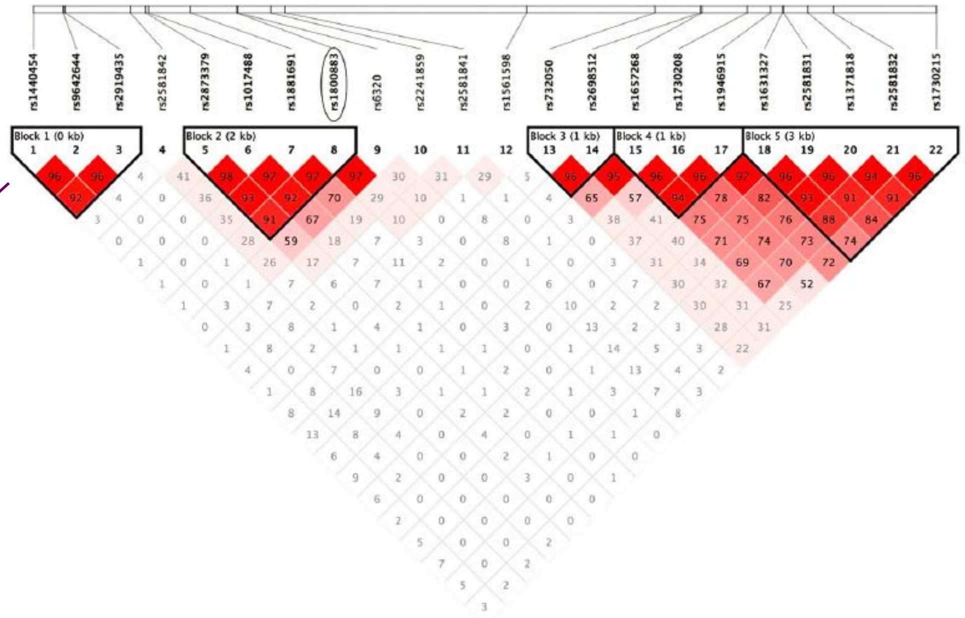
500.000 – 2 million oligonucleotide probes

Q: How do we get information for the rest of the human genome (est. 10 million) ?

= *non-random co-inheritance of nearby variants*

- Based on the linkage disequilibrium (LD) between variants
  - **“Haplotypes” = group of alleles inherited together**

Haplotype



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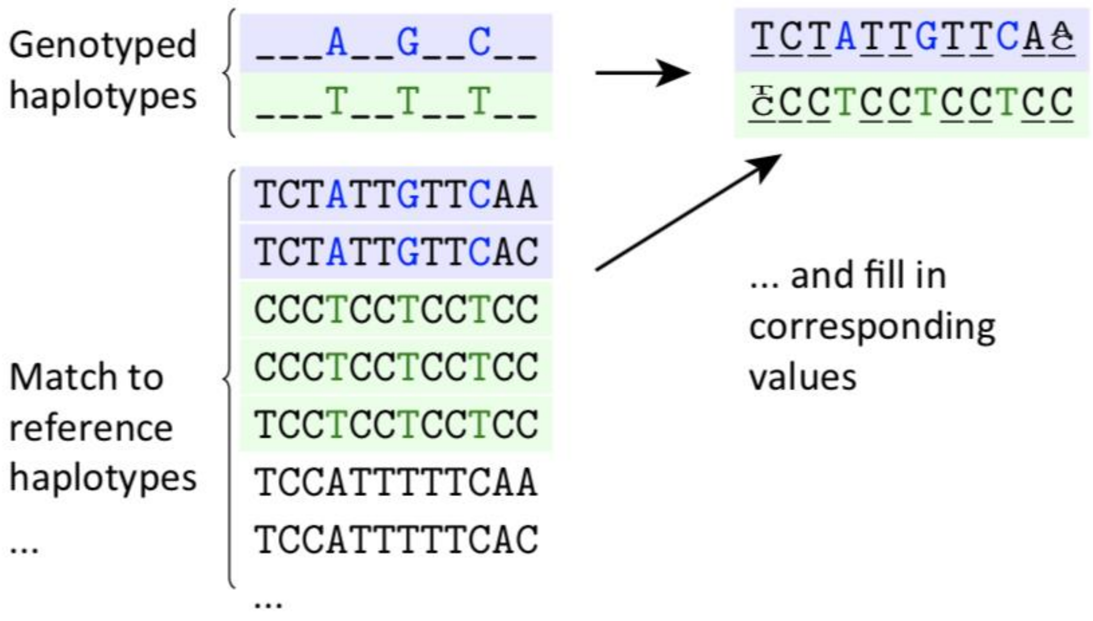
Imputation

- Use the correlations between the SNPs to ‘guess’ the genotypes at untyped positions

- Reference datasets such as www.1000genomes.org, www.uk10k.org and www.haplotype-reference-consortium.org
  - Importance to match the population
  - Panels for non-European populations:

Haplotypes differ between populations

- Example of software used for imputation: + REF



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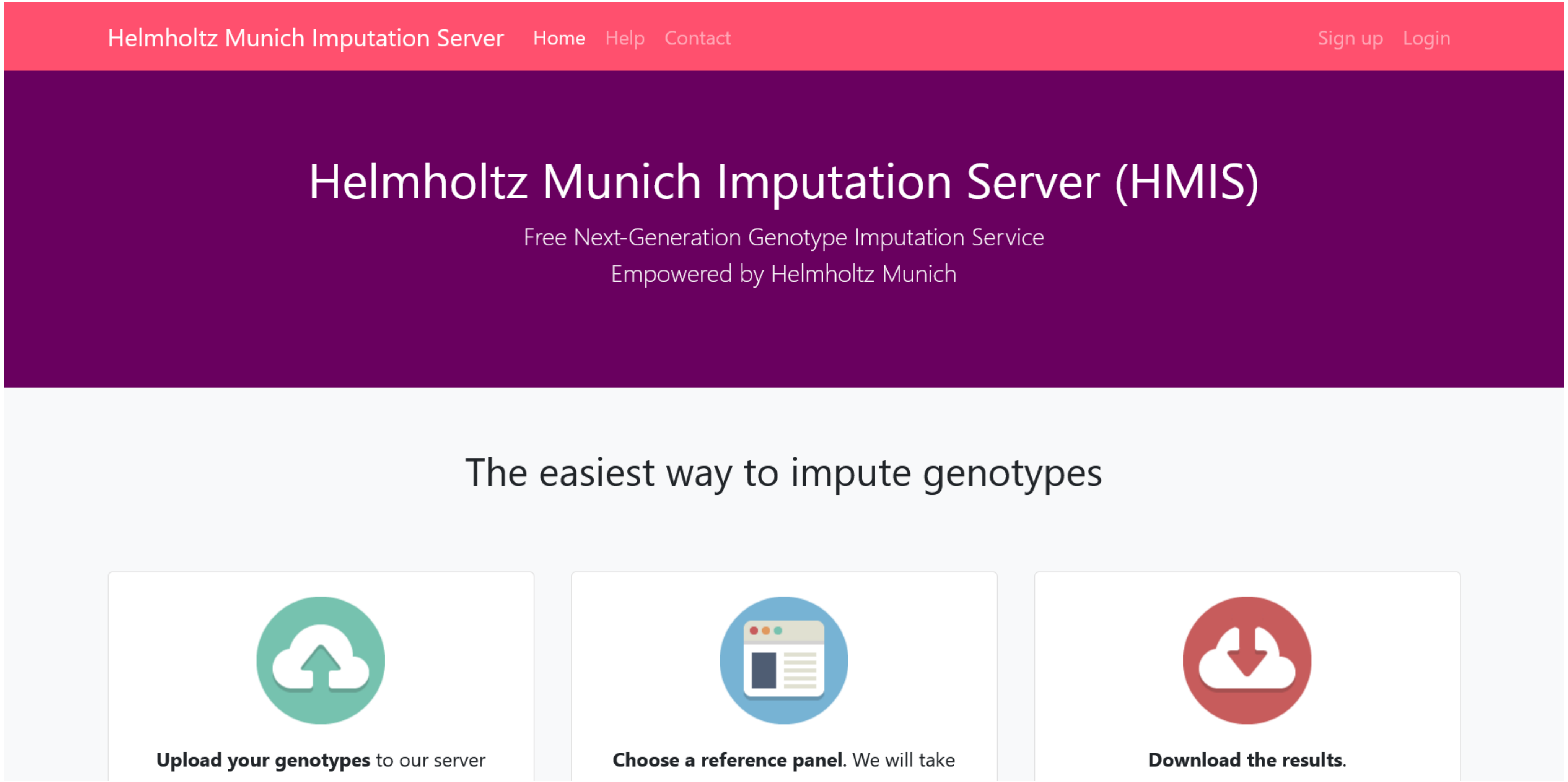
Imputation resources



GDPR compliant service in EU

cost-free

Genotyping arrays

Pros

- Cheap technology
- Easy to do
- Standard pipelines for QC

Cons

- Interrogate only known variations
- Limitations when comparing different studies
- Imputation is limited for rare variants (low LD)

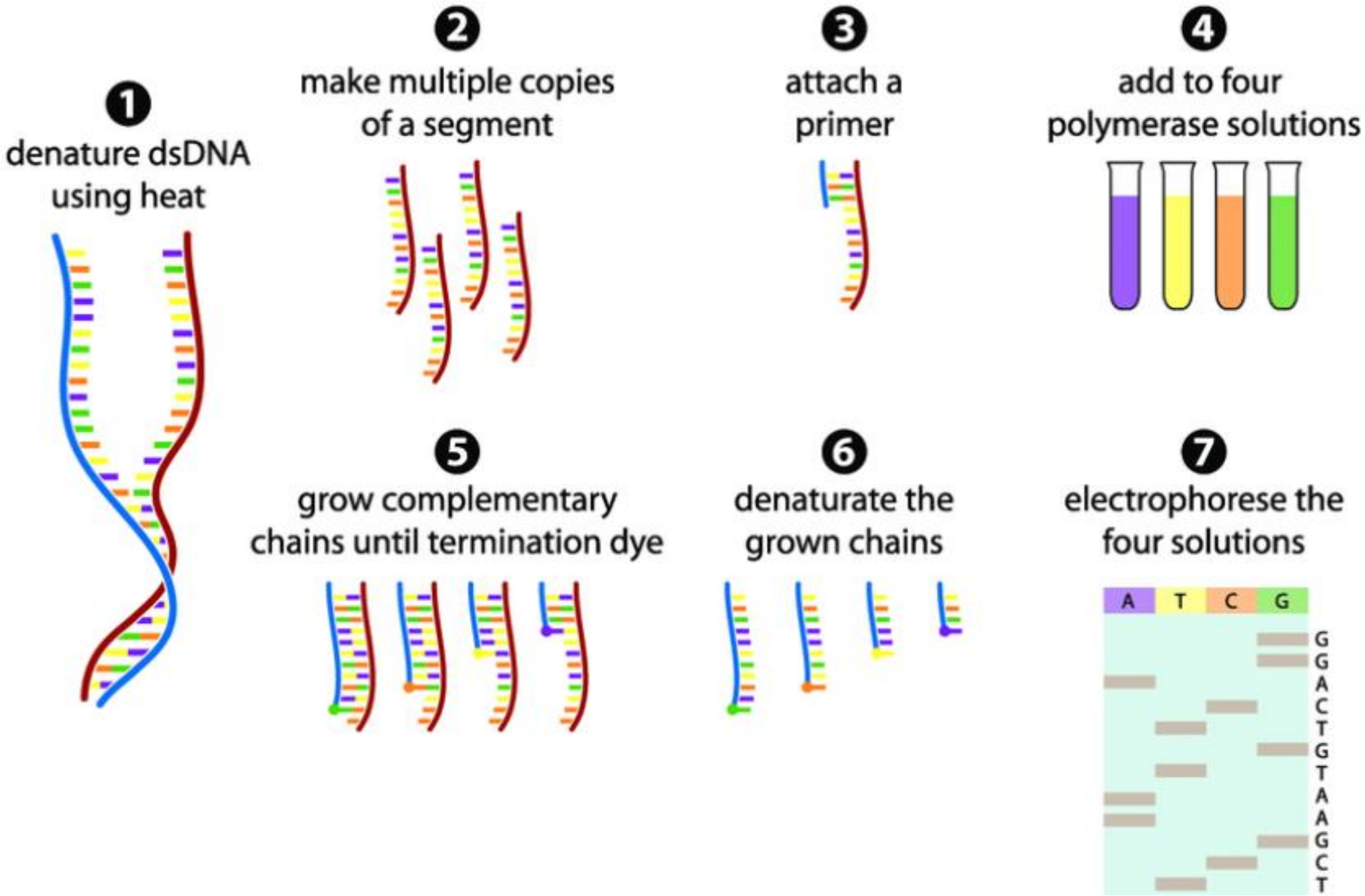
2. Next Generation Sequencing (NGS)

→ Reading the **genome sequence** directly (no fixed probes)

Sequencing = get genotype information for **every position** in the genome including *unknown* variation

Sanger sequencing 1st generation: 1977 –

Reads = consecutive DNA sequence from a fragment (~ 700 bp)



- single-stranded DNA
- DNA-primers (short DNA sequence needed for synthesis initiation)
- deoxynucleotide triphosphates (dNTPs)
- modified di-deoxynucleotide triphosphates (ddNTPs) (terminate DNA strand elongation, labelled)

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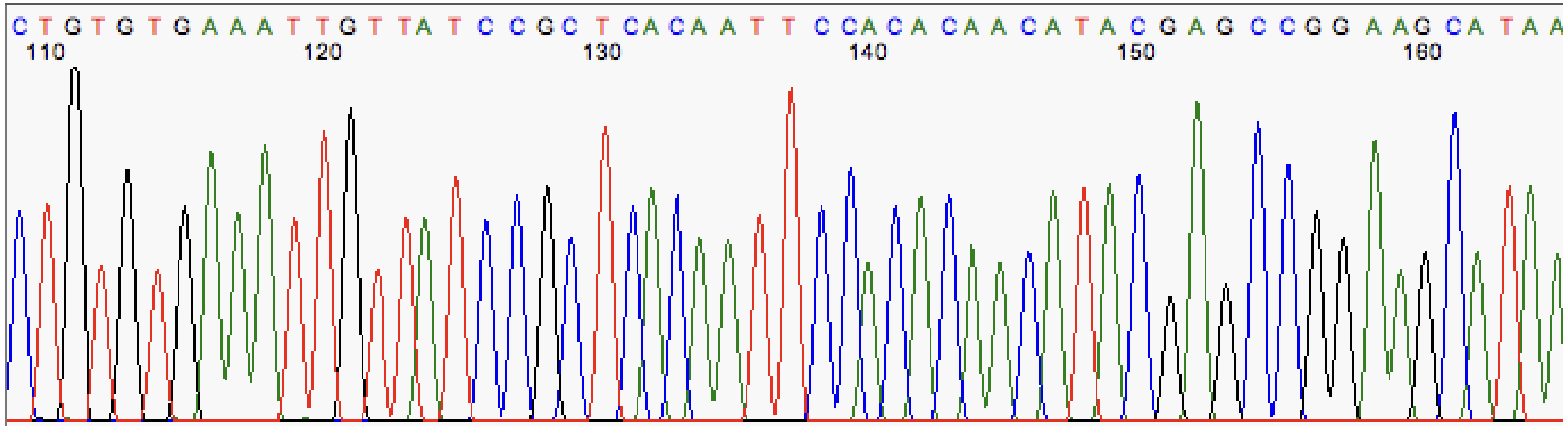
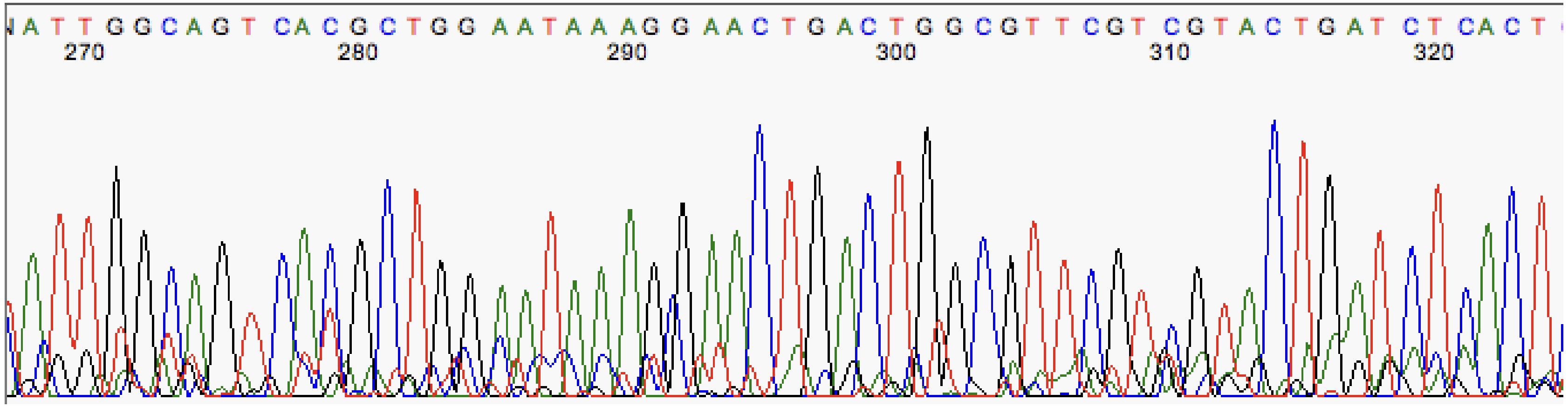
Sanger sequencing

1st generation: 1977 – Reads ~ 700 bp





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Sanger is still the gold standard for VALIDATING a single variant

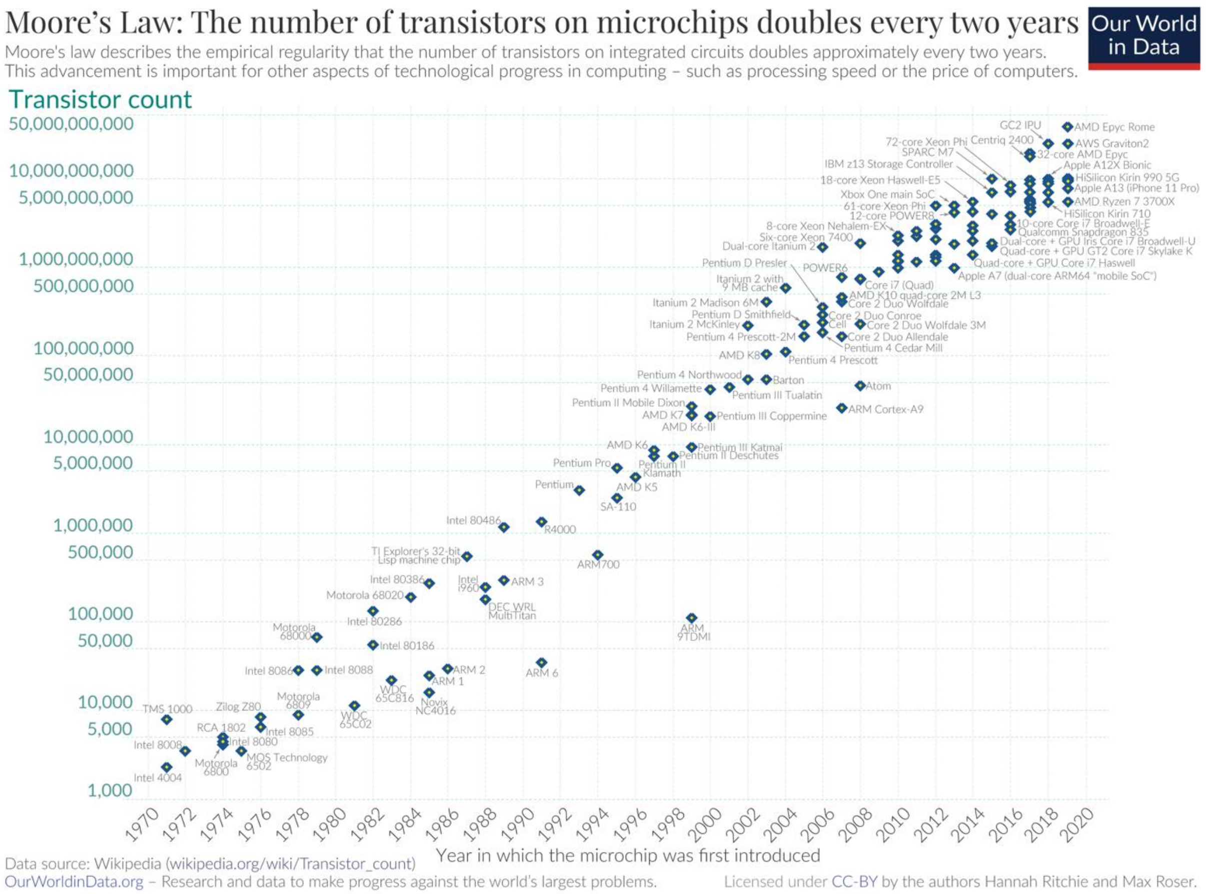
NGS Sequencing –
From sequencing ***one*** *fragment* at a time to ***multiple*** fragments ***in parallel***

- NGS = Next Generation Sequencing
    - More efficient technologies for sequencing, especially through the use of multiplexing
    - Smaller reads

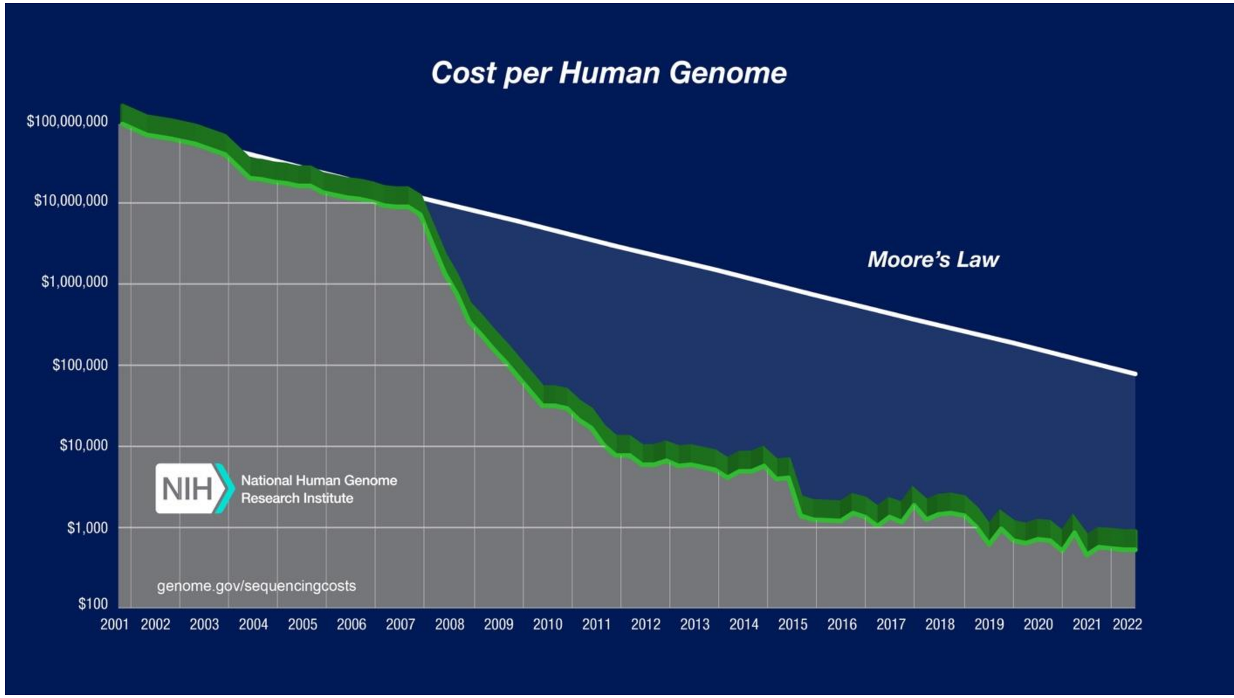
- 2 main technologies:
    - IonTorrent: transistor sequencing - detects pH change
    - **Illumina: Sequencing-By-Synthesis (SBS)**

- Illumina is now(still) the market leader due to very high throughput and low prices

NGS price is getting lower and lower



- First genome: $300 million
- 2008: ~$10 million
- 2014: ~$1,000
- Today: <$200



(doubling of chip density ≈ halving of cost every 2 years).

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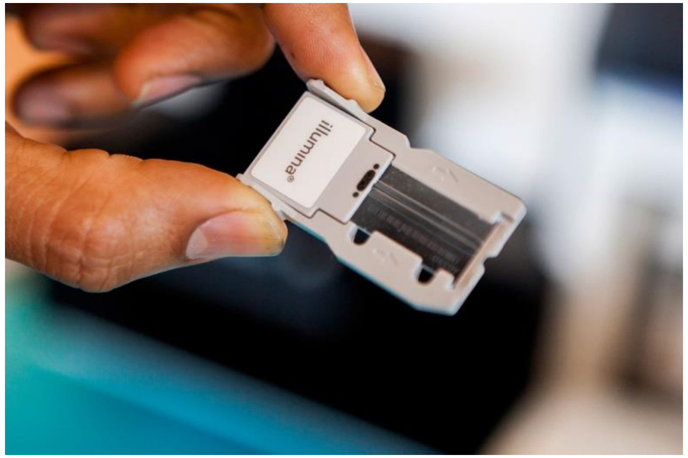
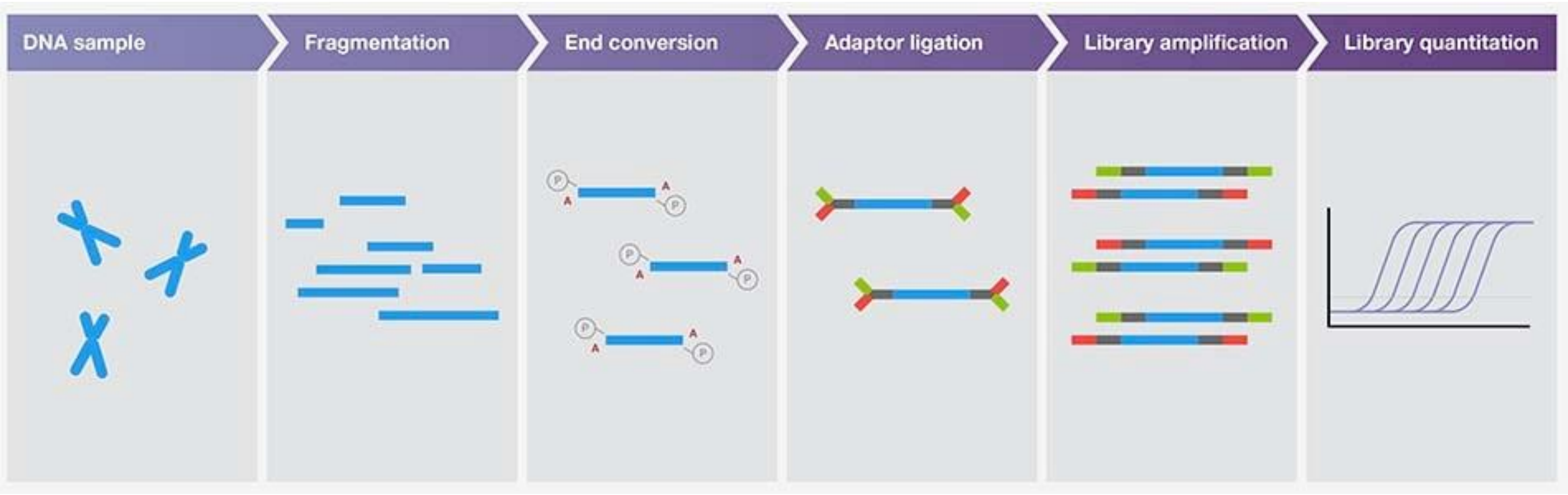
Illumina sequencer

Flow cells



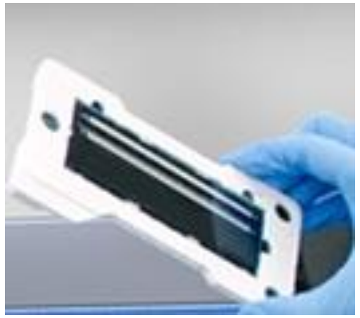
Illumina workflow



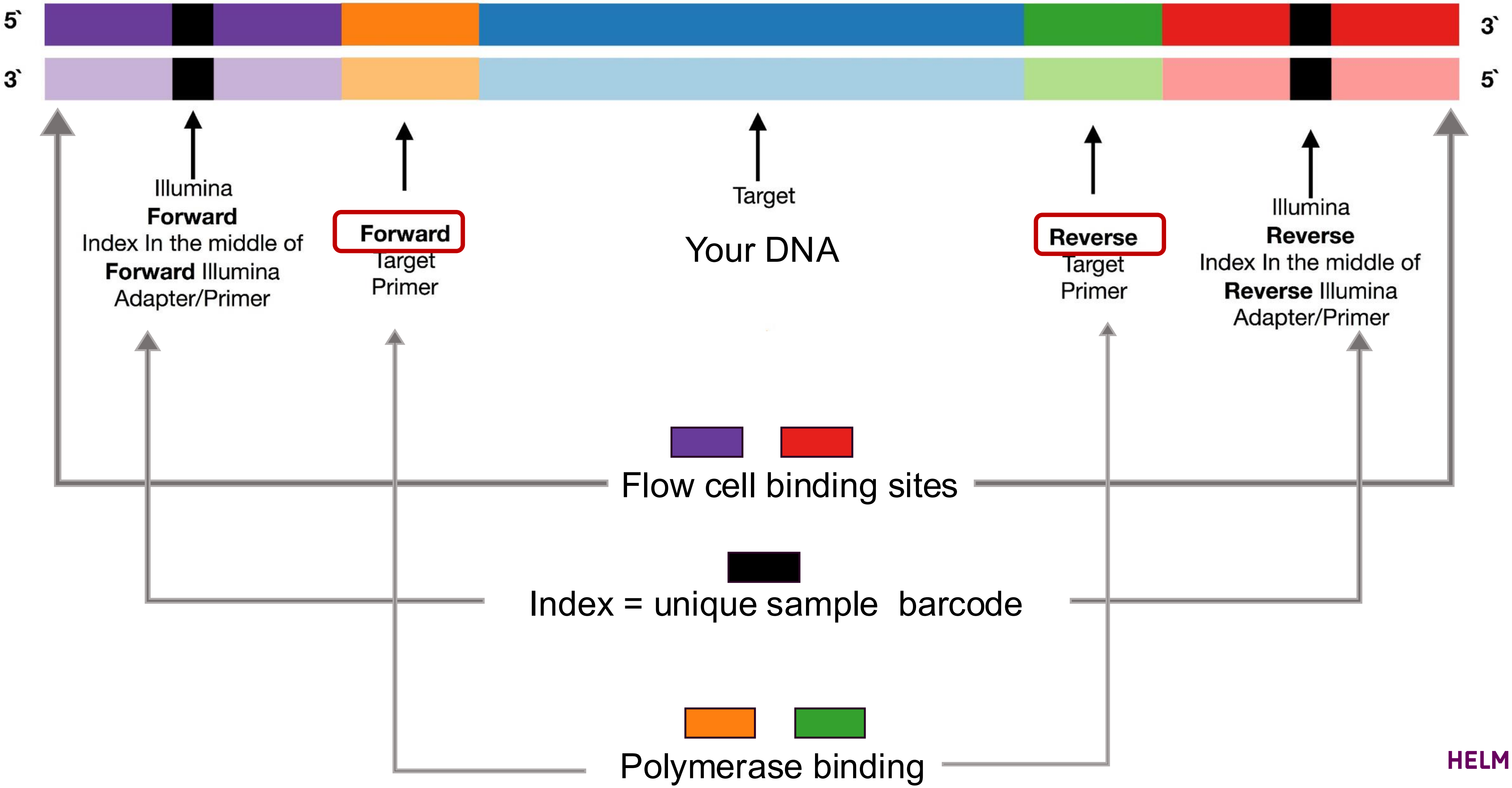
clean-up of messy ends

Adaptors: DNA sequences that carry:

1. Flow cell binding sites
2. Sequencing primer binding sites
3. Sample indexes: (8–10 bp) that identify which sample each read came from (multiplexing)



How does a sequencing read look like?



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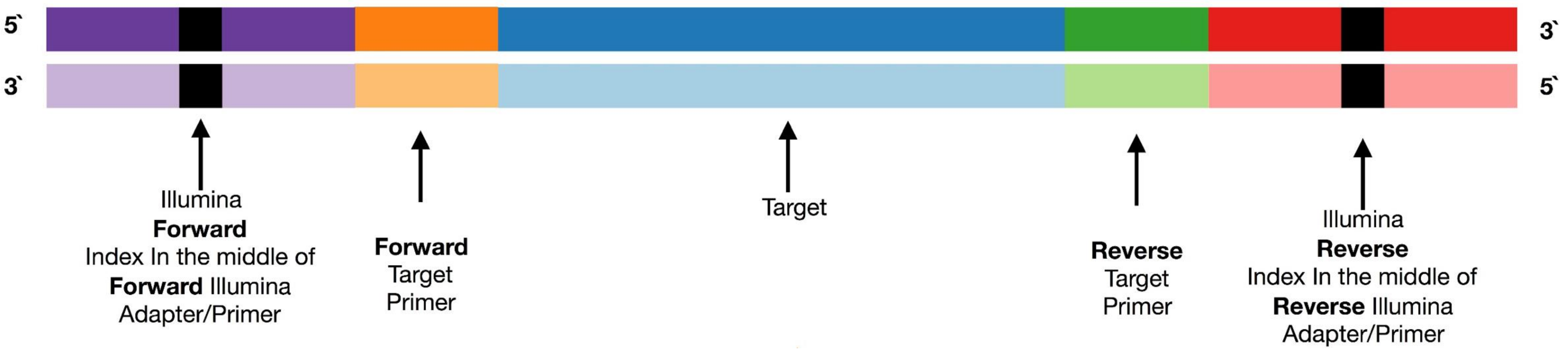
Paired-end sequencing



Two reads from **one fragment: paired-end sequencing**

Fragment (~300 bp):

[==]

← R1 (150 bp) ———┤ ├——— R2 (150 bp) →

[unsequenced gap]

← read this end read this end →

Why paired-end?

1. Better alignment
2. Detect structural changes
3. Better quality coverage

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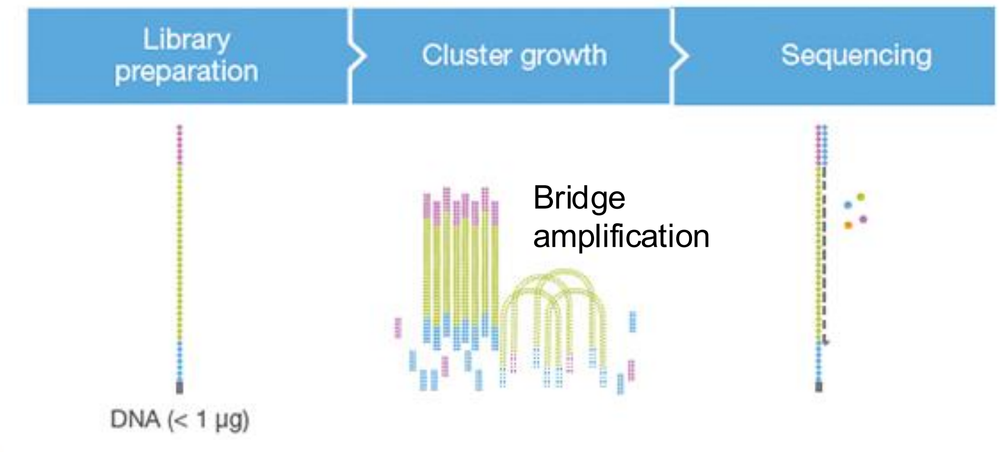
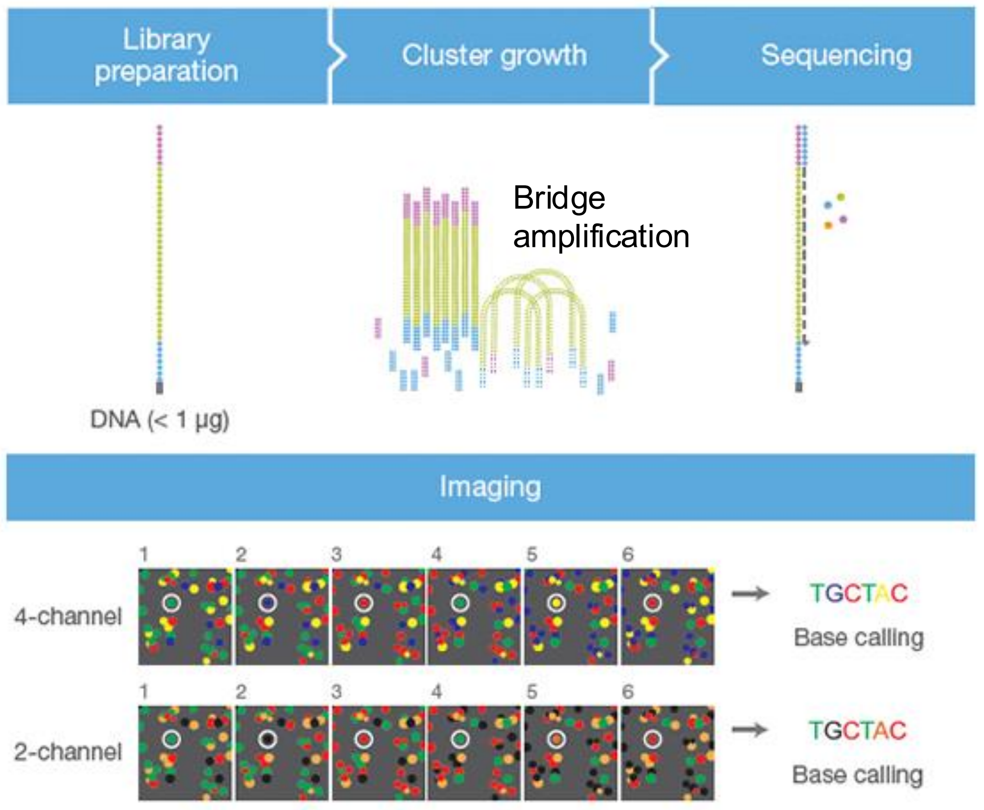
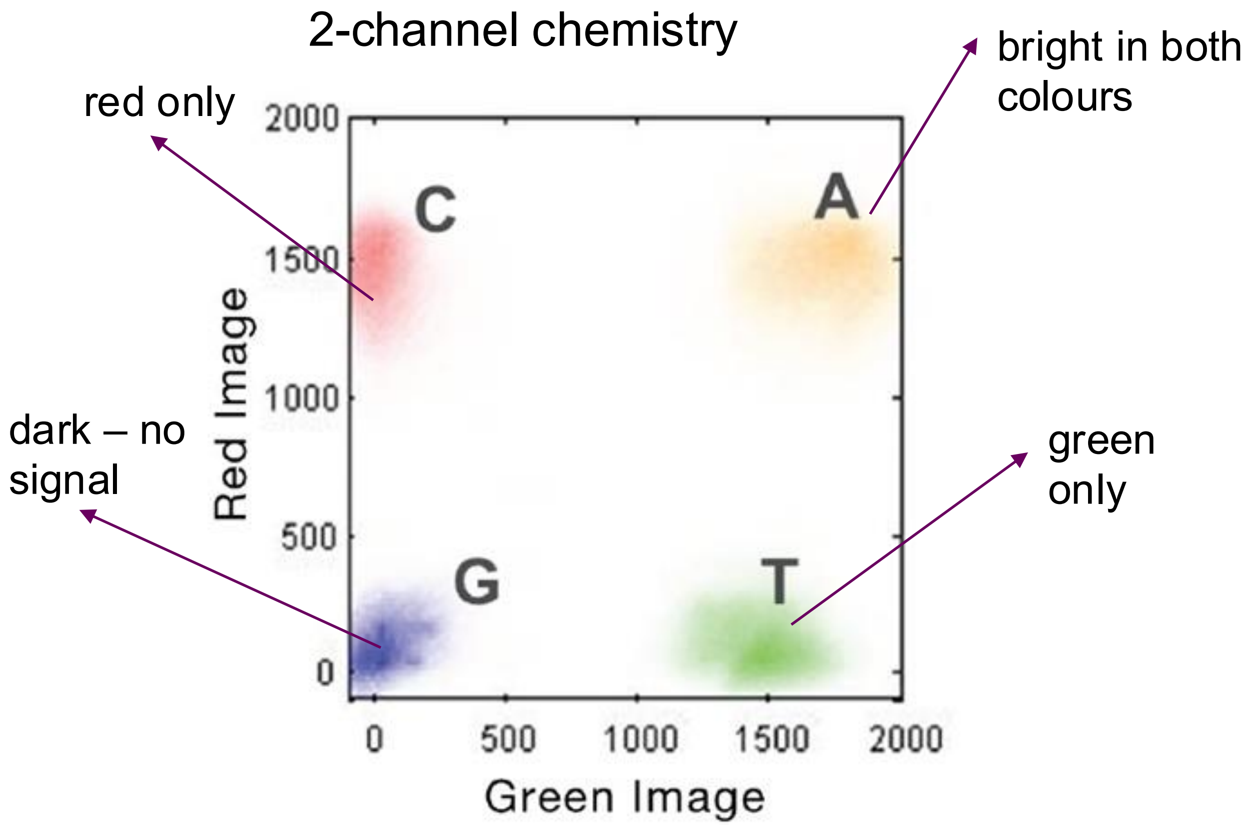
Illumina Sequencing By Synthesis (SBS)



cluster of approx. 1,000 identical copies of the same fragment

Illumina Sequencing By Synthesis (SBS)

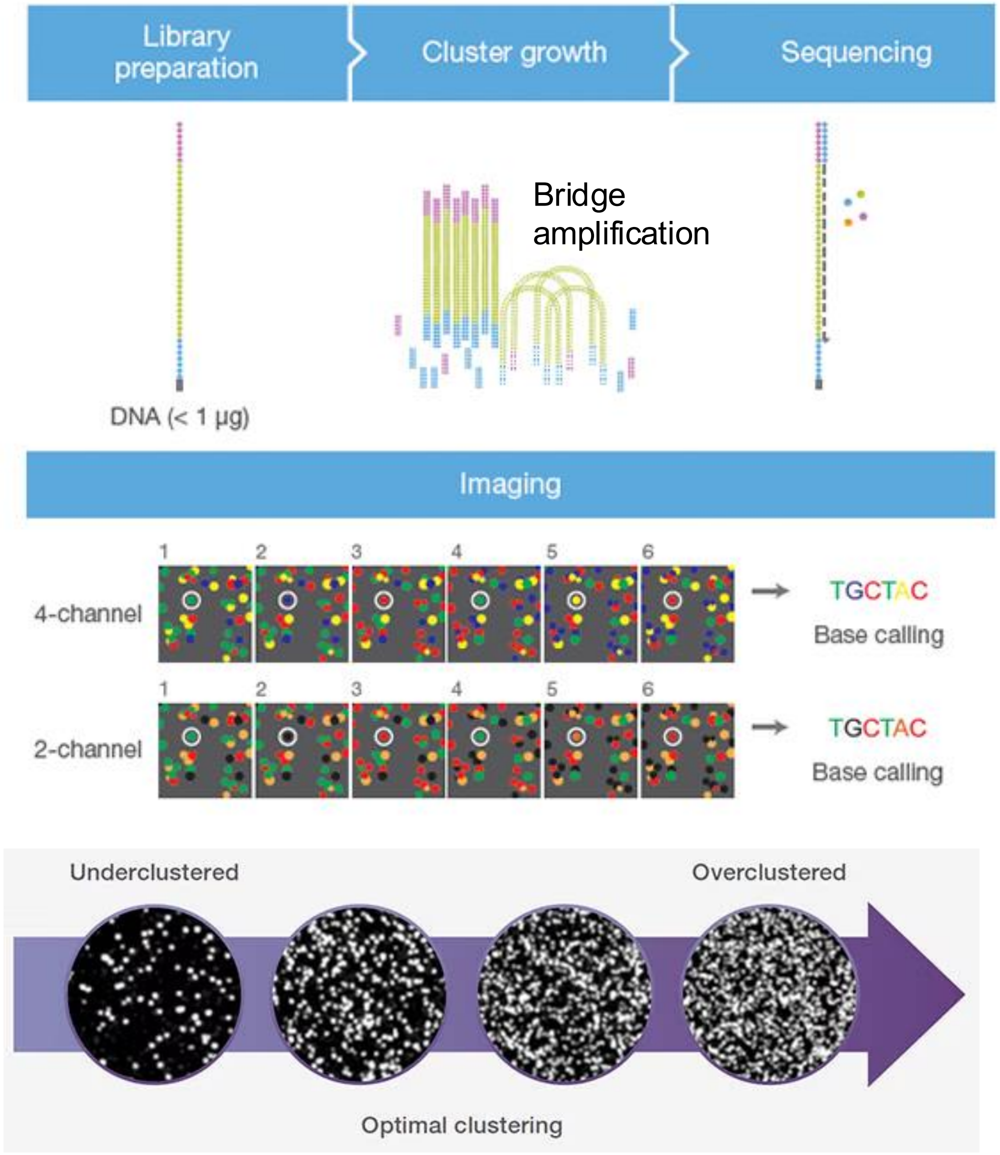




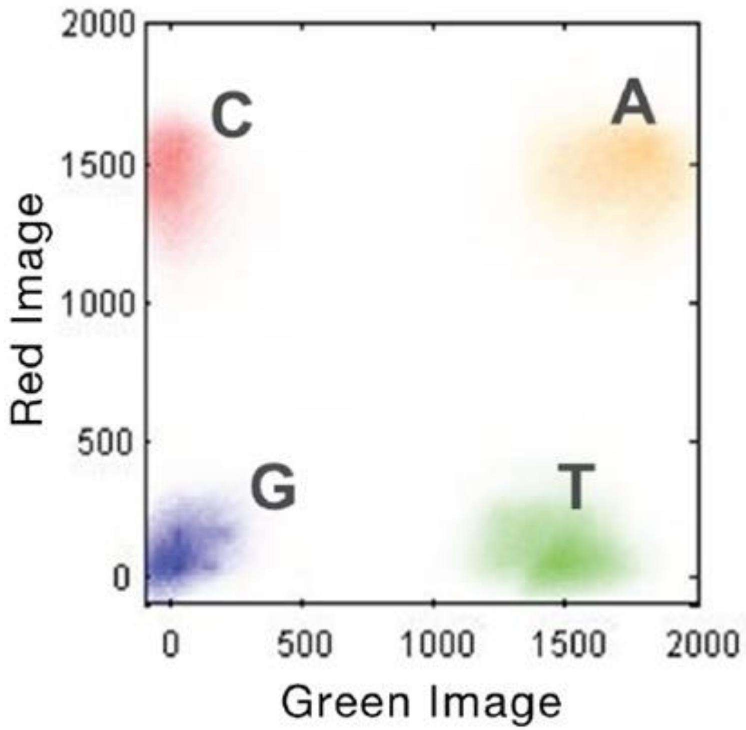
Each imaging cycle (addition of 1 base) has 4 steps:

1. **Add** all four nucleotides (fluorescent + chemical block)
2. **Wash** away the extras
3. **Image** (take a picture)
4. **Cleave** the dye and the block

Illumina Sequencing By Synthesis (SBS)



2-channel chemistry



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FASTQ format

Instrument Name | Flowcell Info | x,y of cluster in tile | index #, member of pair

```
@H06HDADXX130110:2:2116:3345:91806/1
GTTAGGGTTAGGGTTGGGTTAGGGTTAGGGTTAGGGTTAGGGGTAGGG....
+
>=<=???>?>???=??>>8<?><=2=<===1194<?;:?>>?#3==>##...
```

Raw sequence letters

Quality values of raw sequence letters

Phred quality scores

How confident is the machine about each base?

FASTQ format

Instrument Name | Flowcell Info | x,y of cluster in tile | index #, member of pair

```
@H06HDADXX130110:2:2116:3345:91806/1
GTTAGGGTTAGGGTTGGGTTAGGGTTAGGGTTAGGGTTAGGGGTAGGG....
+
>=<=???>?>???=??>>8<?><=2=<===1194<?;:?>>?#3==>##...
```

Raw sequence letters

Quality values of raw sequence letters

Table 1 ASCII Characters Encoding Q-scores 0–40

Symbol	ASCII Code	Q-Score	Symbol	ASCII Code	Q-Score	Symbol	ASCII Code	Q-Score
!	33	0	/	47	14	=	61	28
"	34	1	0	48	15	>	62	29
#	35	2	1	49	16	?	63	30
$	36	3	2	50	17	@	64	31
%	37	4	3	51	18	A	65	32
&	38	5	4	52	19	B	66	33
'	39	6	5	53	20	C	67	34
(	40	7	6	54	21	D	68	35
)	41	8	7	55	22	E	69	36
*	42	9	8	56	23	F	70	37
+	43	10	9	57	24	G	71	38
,	44	11	:	58	25	H	72	39
-	45	12	;	59	26	I	73	40
.	46	13	<	60	27			

Q-score	Error probability
Q10	1:10 (10%)
Q20	1:100 (1%)
Q30	1:1000 (0.1%)
Q40	1:10.000 (0.01%)

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FASTQ format

Instrument Name | Flowcell Info | x,y of cluster in tile | index #, member of pair

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@H06HDADXX130110:2:2116:3345:91806/1
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,	44	11	:	58	25	H	72	39
-	45	12	;	59	26	I	73	40
.	46	13	<	60	27			

Q-score	Error probability
Q10	1:10 (10%)
Q20	1:100 (1%)
Q30	1:1000 (0.1%)
Q40	1:10.000 (0.01%)

Q30: 99.9% that the base call is correct

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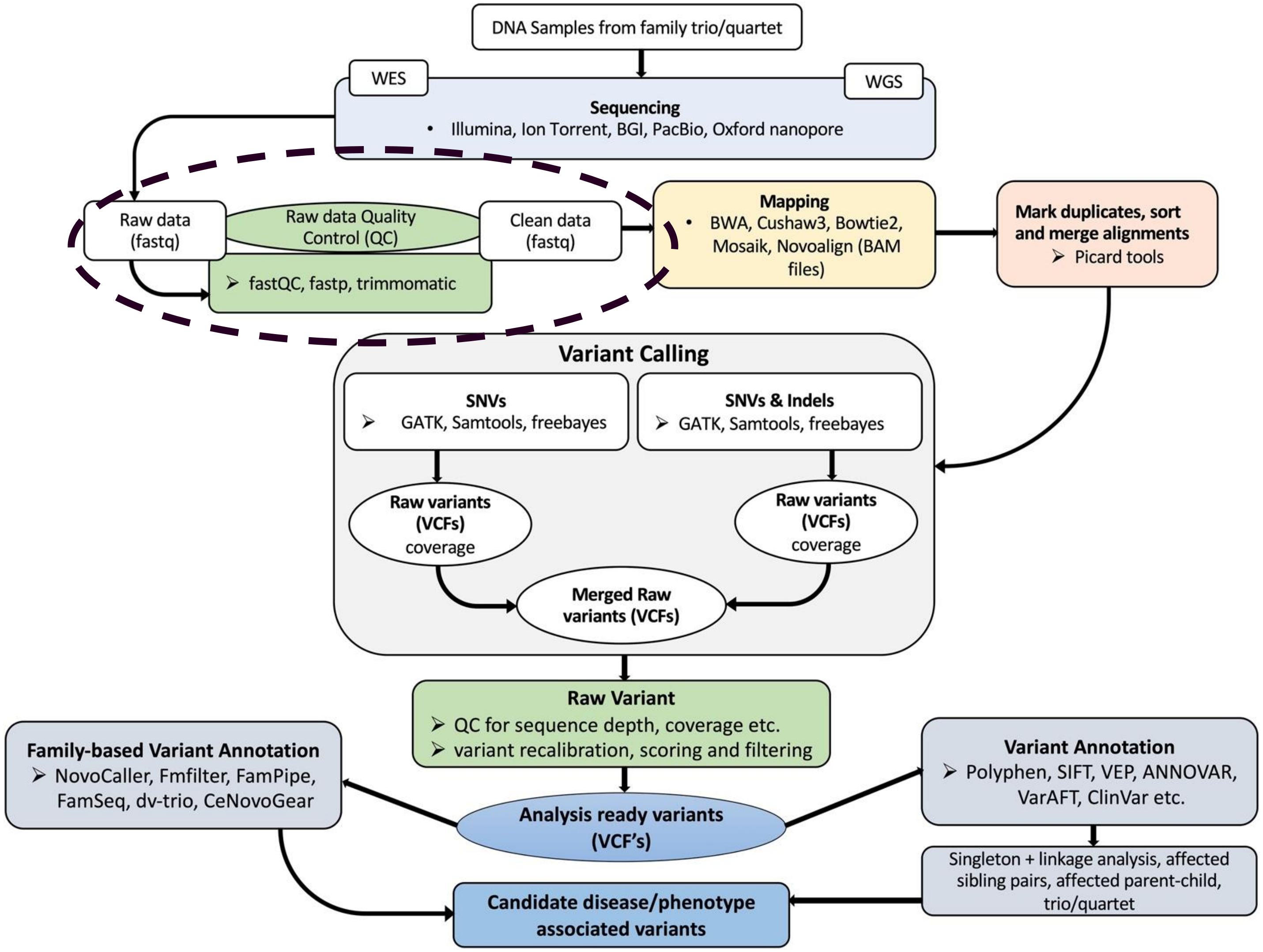
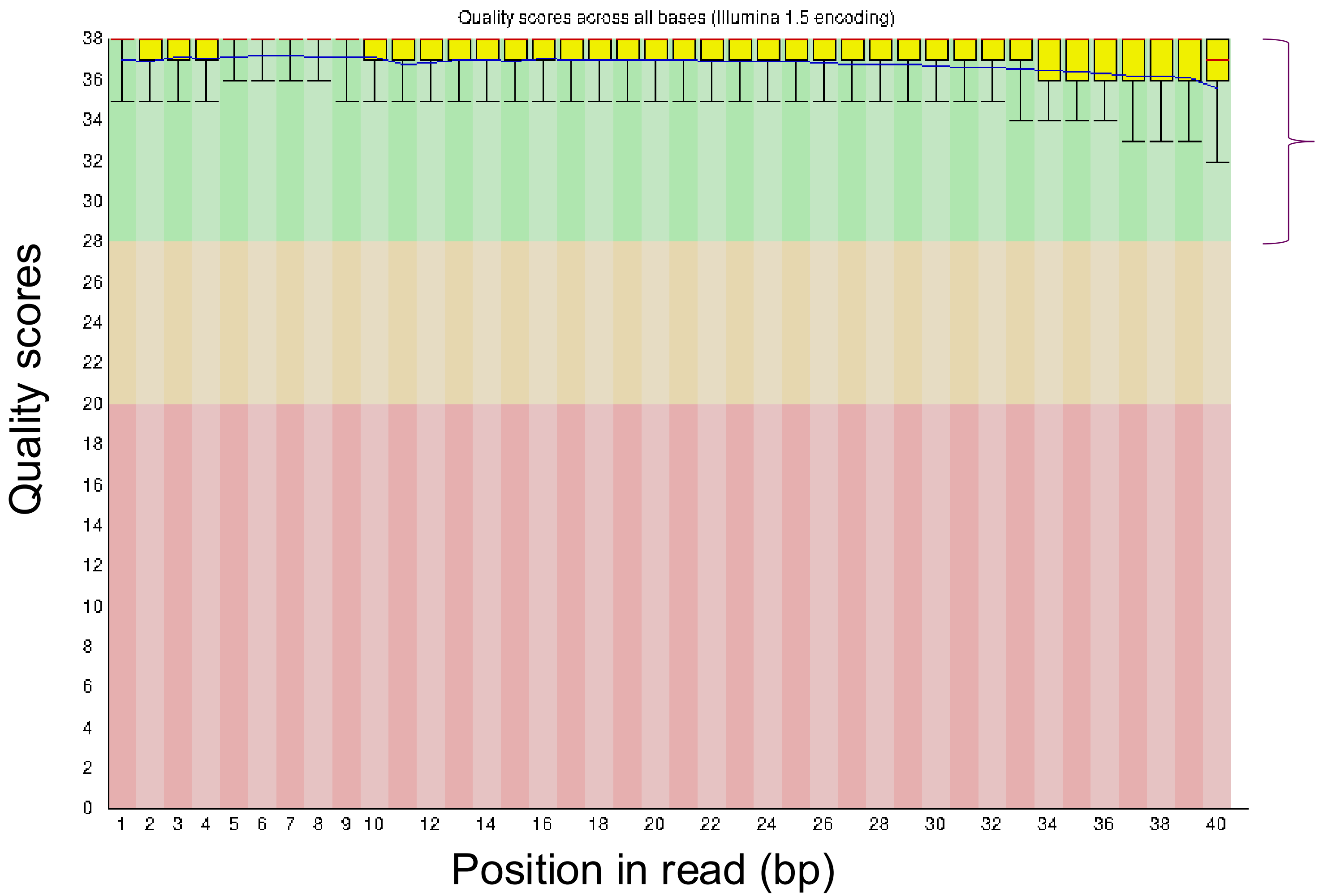
NGS variant calling bioinformatic pipeline



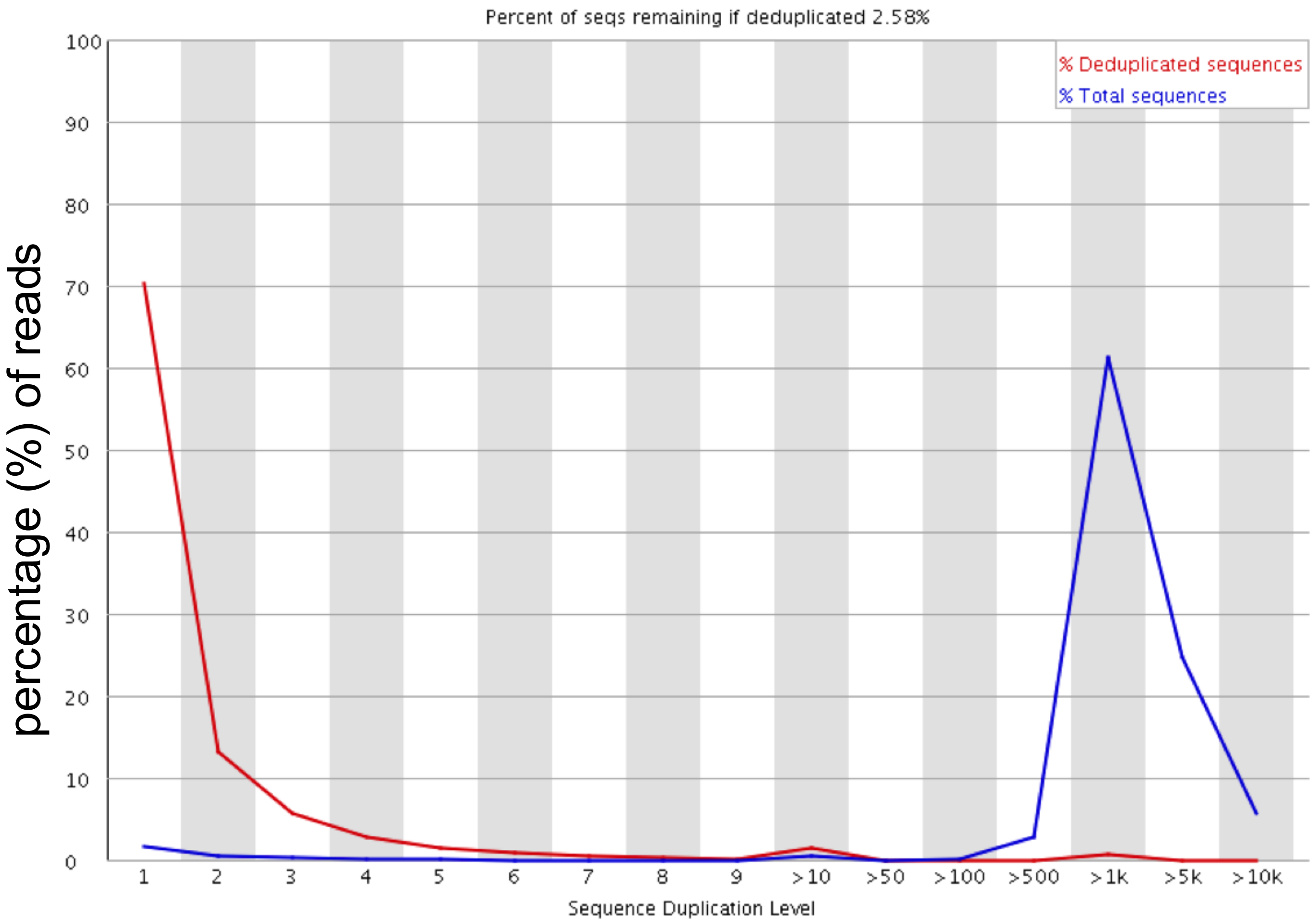
https://doi.org/10.3389/fgene.2020.544162

FASTQ QC - Per-base quality

Per-base quality



FASTQ QC - Sequence Duplication Levels



% deduplicated sequences - diversity of unique sequences.

% Total sequences - actual composition.

PCR over-amplification due to low input DNA

duplication level - how many times an identical sequence appears in the data

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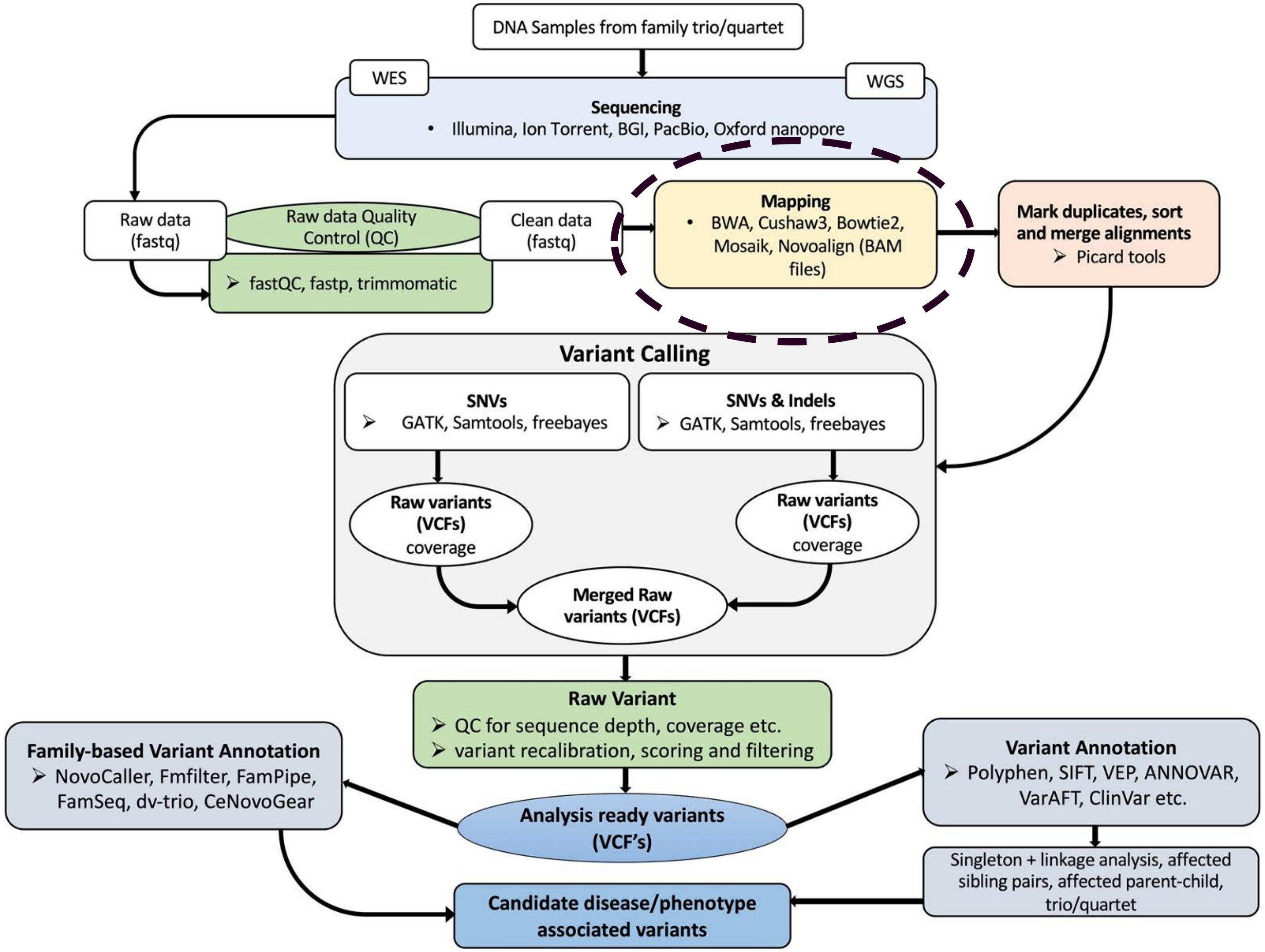
NGS variant calling bioinformatic pipeline



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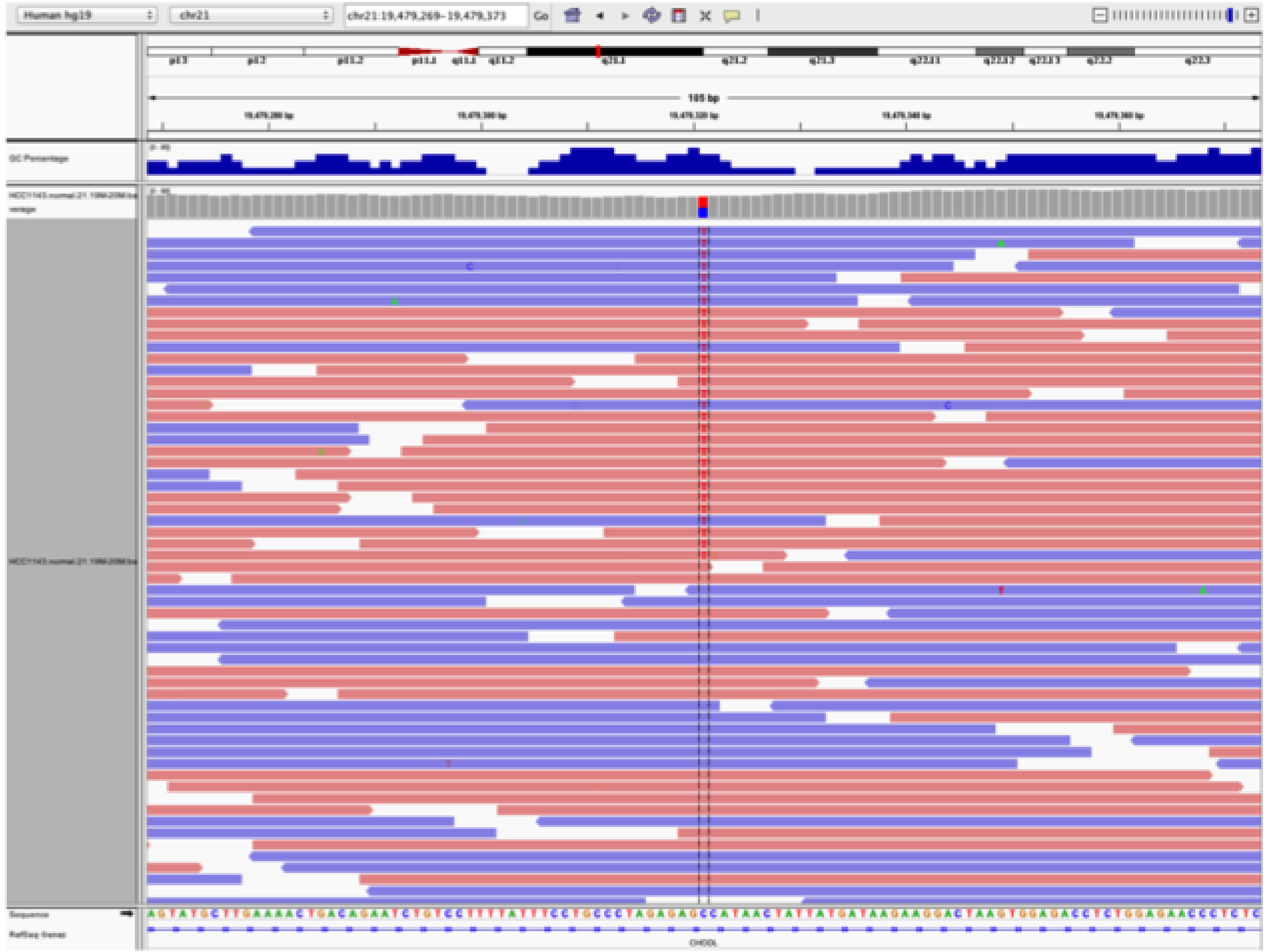
https://doi.org/10.3389/fgene.2020.544162

Alignment or assembly

Aligned reads



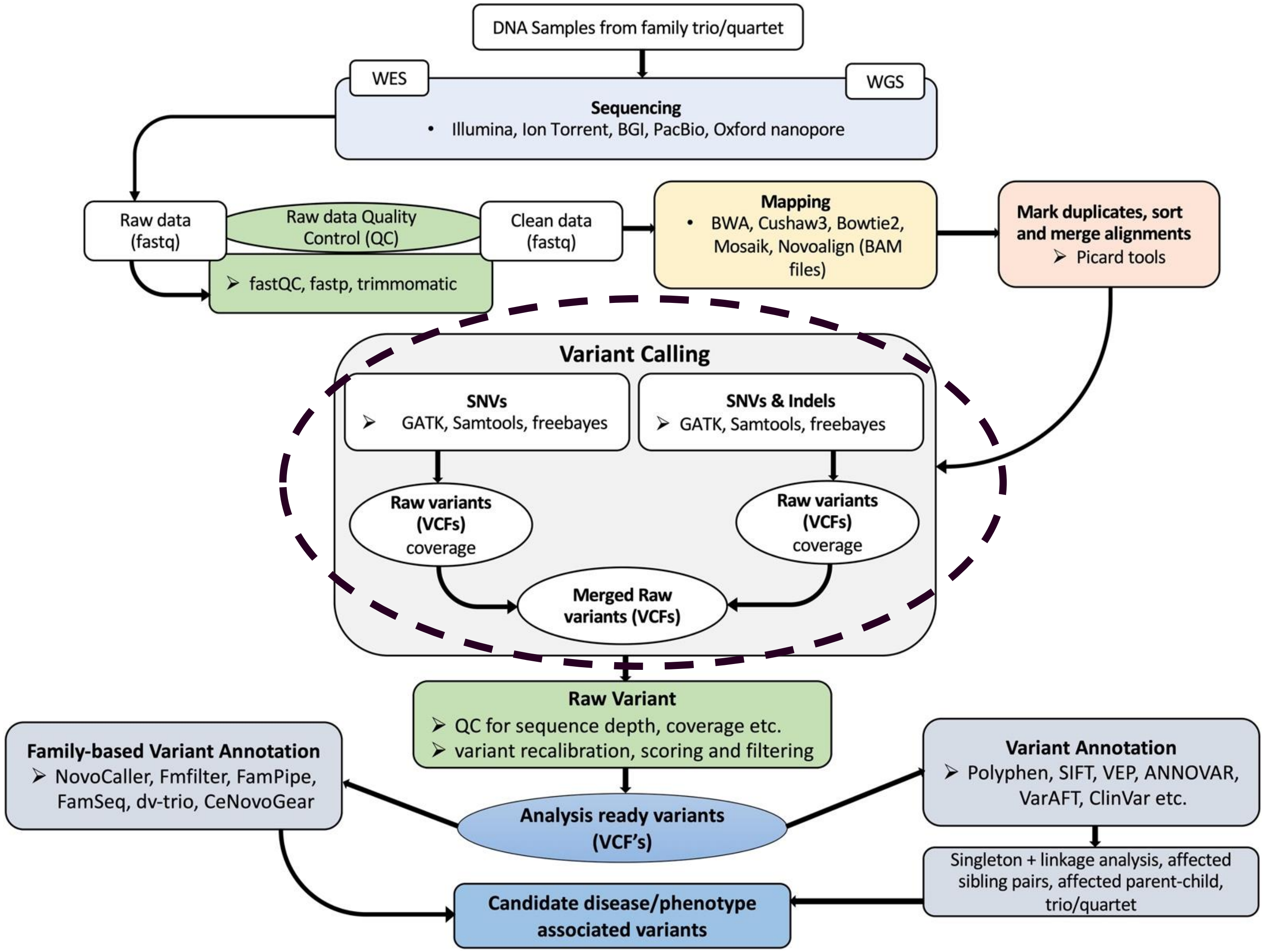
Reference SNP

- Need to know where the reads map to the genome
- Takes into account read orientation and sequence similarity
- If the read does not match entirely: is it at the wrong place? A sequencing error? A SNP?

  → Probabilistic process → MAPQ
- Alignment requires a reference genome to align to

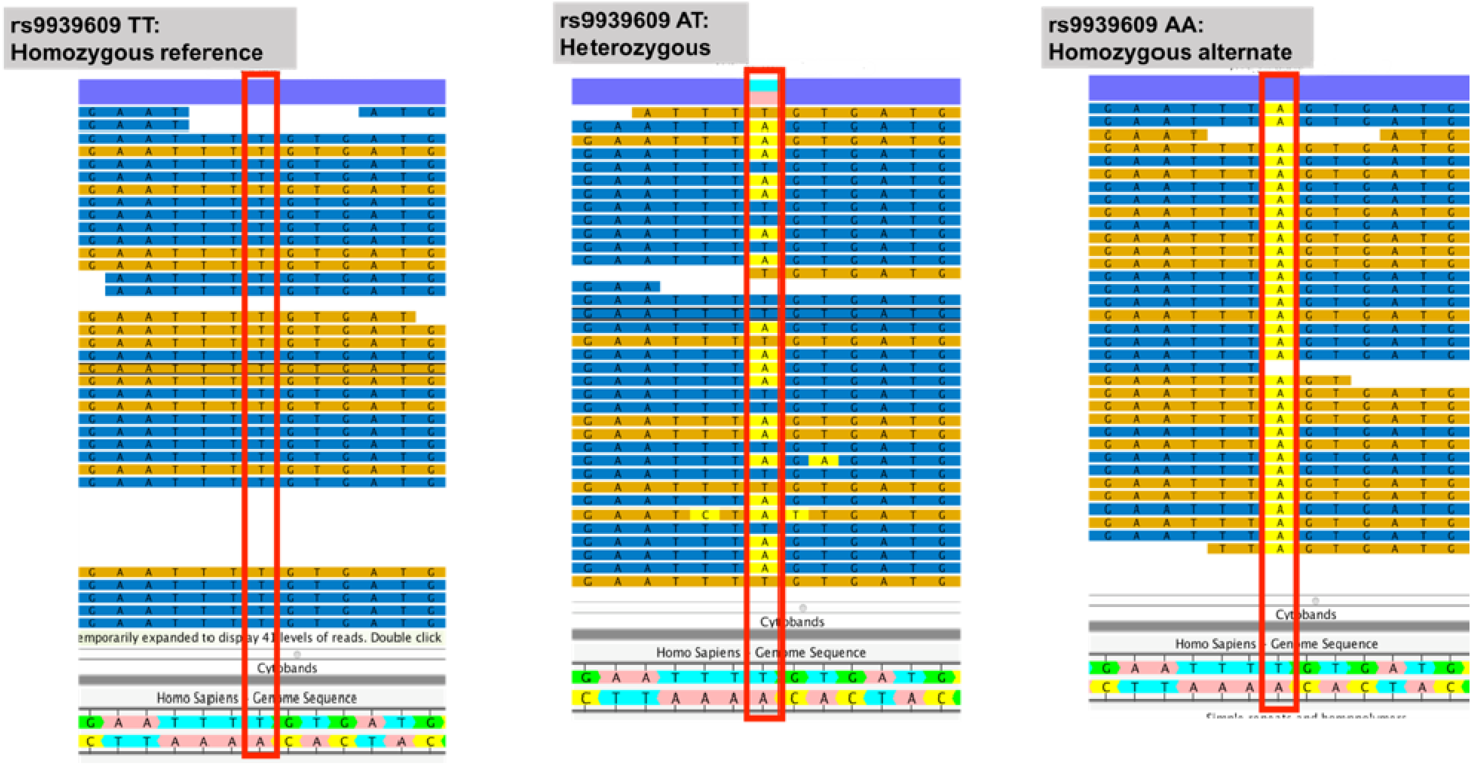
  → Available for humans but not for all the species
- Example of software: BWA, Bowtie2

NGS variant calling bioinformatic pipeline



https://doi.org/10.3389/fgene.2020.544162

Variant calling



- Step to determine the genotypes of the individuals
- Output format = VCF which will be used for association analyses
- Quality measures:
  - QualByDepth (QD)
  - FisherStrand (FS)
  - StrandOddsRatio (SOR)
  - RMSMappingQuality (MQ)
  - MappingQualityRankSumTest (MQRankSum)
  - ReadPosRankSumTest (ReadPosRankSum)
- Variant Quality Score Recalibration (VQSR) – ML to flag poor quality variants
- Measure of overall quality → Ti/Tv

VCF file format (**V**ariant **C**all **F**ormat)

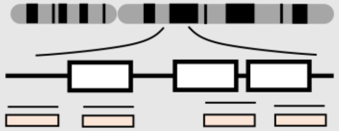
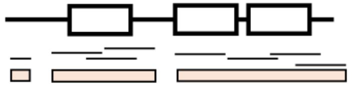
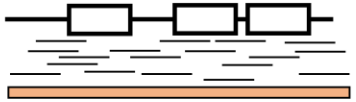
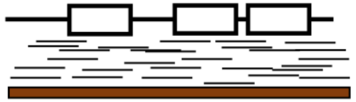
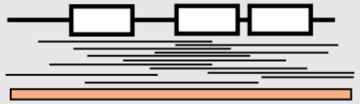
Header

```
##fileformat=VCFv4.1
##fileDate=20110413
##source=VCFtools
##reference=file:///refs/human_NCBI36.fasta
##contig=<ID=1,length=249250621,md5=1b22b98cdeb4a9304cb5d48026a85128,species="Homo Sapiens">
##contig=<ID=X,length=155270560,md5=7e0e2e580297b7764e31dbc80c2540dd,species="Homo Sapiens">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
#CHROM POS ID     REF  ALT    QUAL FILTER  INFO                 FORMAT    SAMPLE1  SAMPLE2
```

Body

```
1        1  .     ACG  A,AT    40  PASS    .                    GT:DP     1/1:13   2/2:29
1        2  .     C    T,CT    .   PASS    H2;AA=T              GT        0|1      2/2
1        5  rs12  A    G       67  PASS    .                    GT:DP     1|0:16   2/2:20
X      100  .     T    <DEL>   .   PASS    SVTYPE=DEL;END=299   GT:GQ:DP  1:12:.   0/0:20:36
```

Sequencing approaches

Sequencing Strategy		Pros	Cons
	Ultra low depth <0.5x	Can use off-target reads from exome sequencing to provide frequencies in understudied populations	Can't reliably provide individual-level genotypes
	Very low depth 0.5-2x	Works best in related samples	Heavy computational requirements
	Low depth 2-8x	Can provide haplotypes for imputation and variant-site discovery	Requires computing and imputation to estimate individual-level calls
	Medium depth (8-25x)	Reasonably high genotype accuracy and capture of rare variants relative to cost	Incomplete capture of indels and very rare variants (e.g., singletons)
	Exome (>20x at target)	Highly accurate genotype calls in covered coding regions - the most directly interpretable portion of the genome	Little coverage outside of coding regions, small fraction of genes missed, harder to call indels and structural variants. Some challenges comparing across capture methods
	High depth (>25x)	Best coverage of sequencable genome, rare variants, and high accuracy of genotype calls	More expensive, doesn't detect structural variants (as previous approaches)
	Long-read	Detection of structural variants, better assembly, mapping and phasing	Data processing, most expensive approach

Box 1: Overview of the different sequencing techniques currently available. White boxes correspond to coding exons and thin black lines to sequencing reads. The sequencing depth is represented at the bottom of each graphic by brown shades. Pros and cons of the different depths and genome coverage are highlighted.

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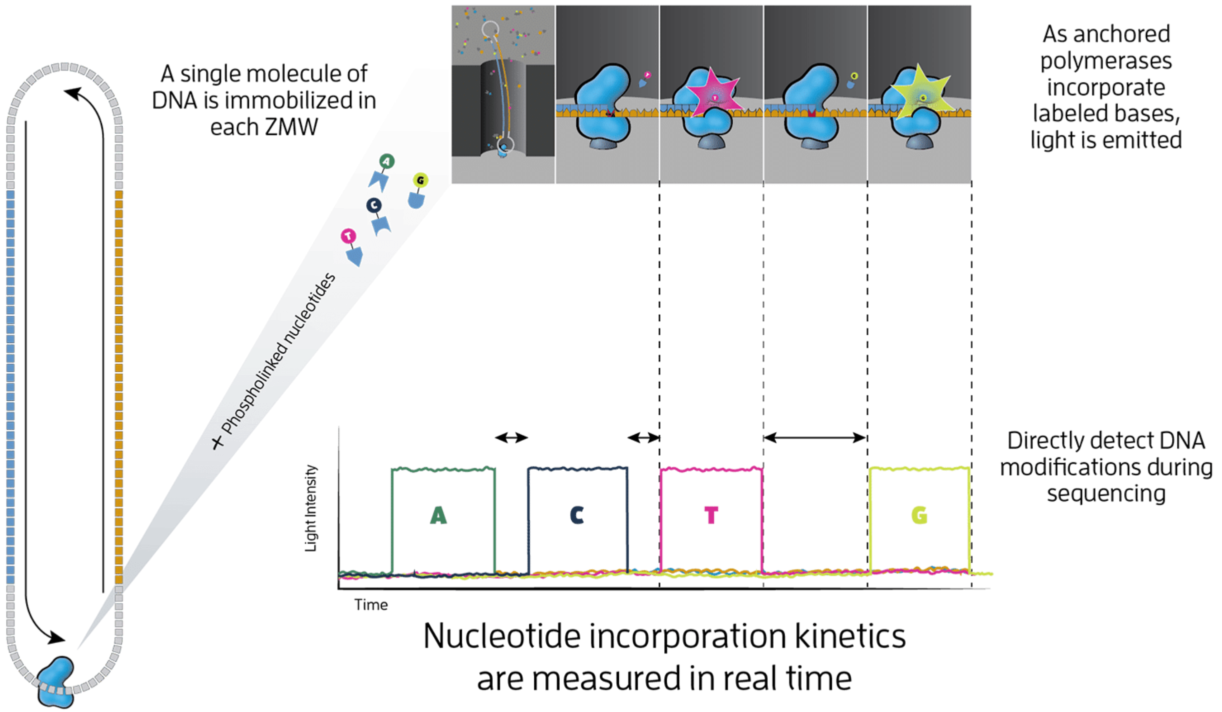
Bocher et al., 2023, Nat Comm

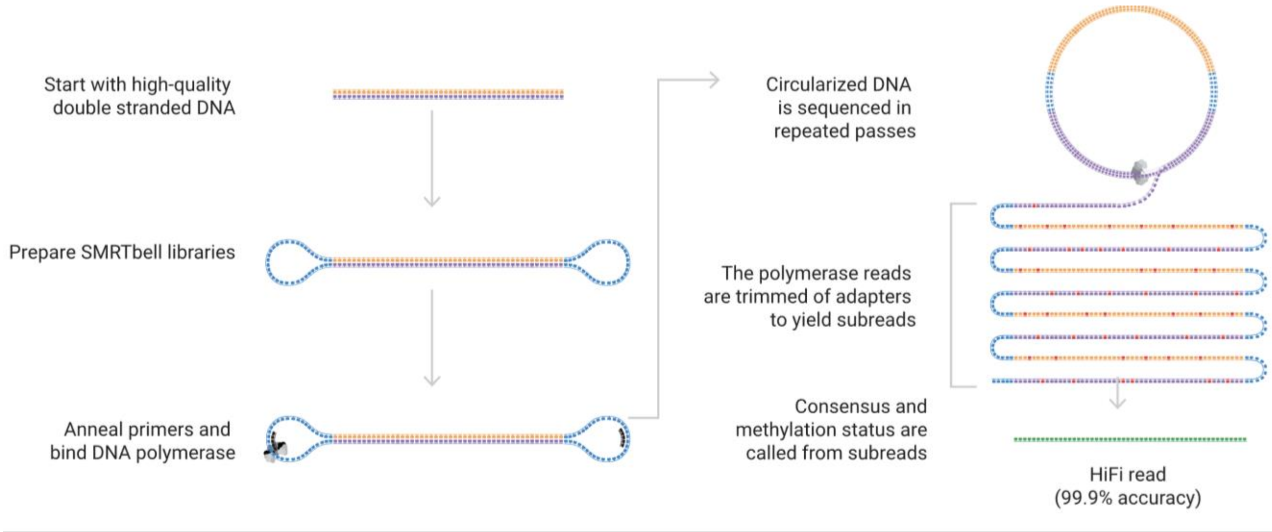
3

Third-generation sequencing

Pacific Biosciences (PacBio): Long-read sequencing

- Reads are > 1kb long
- Accuracy is > 99.9%, approaching Illumina and Sanger sequencing quality
- Particularly suited for *de novo* genome assembly or difficult to map regions





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Nanopore

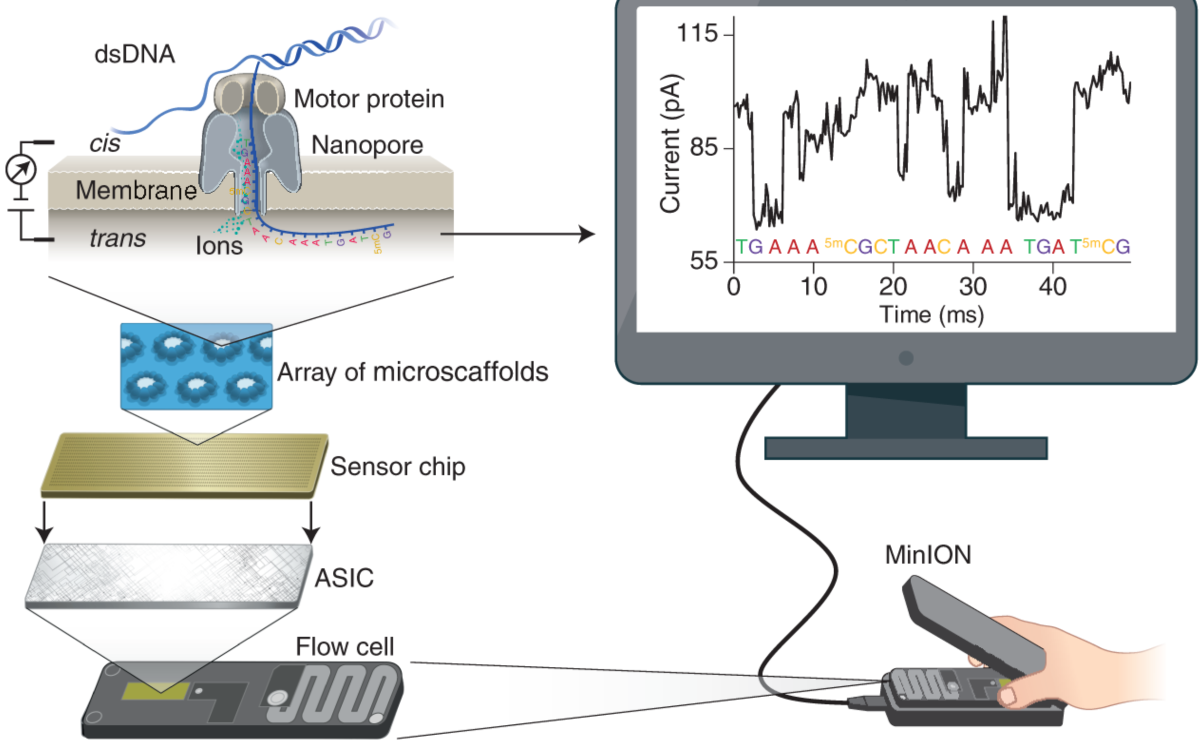
As a nucleotide passes through the pore, it disrupts a current that has been applied to the nanopore. Each nucleotide provides a characteristic electronic signal that is recorded as a current disruption event.

Particularly suited to sequence environmental and metagenomic samples, and repetitive sequences.

Also, now used to study RNA splicing and DNA methylation

Allows for real time sequencing – very portable device

protein nanopore
embedded in a membrane

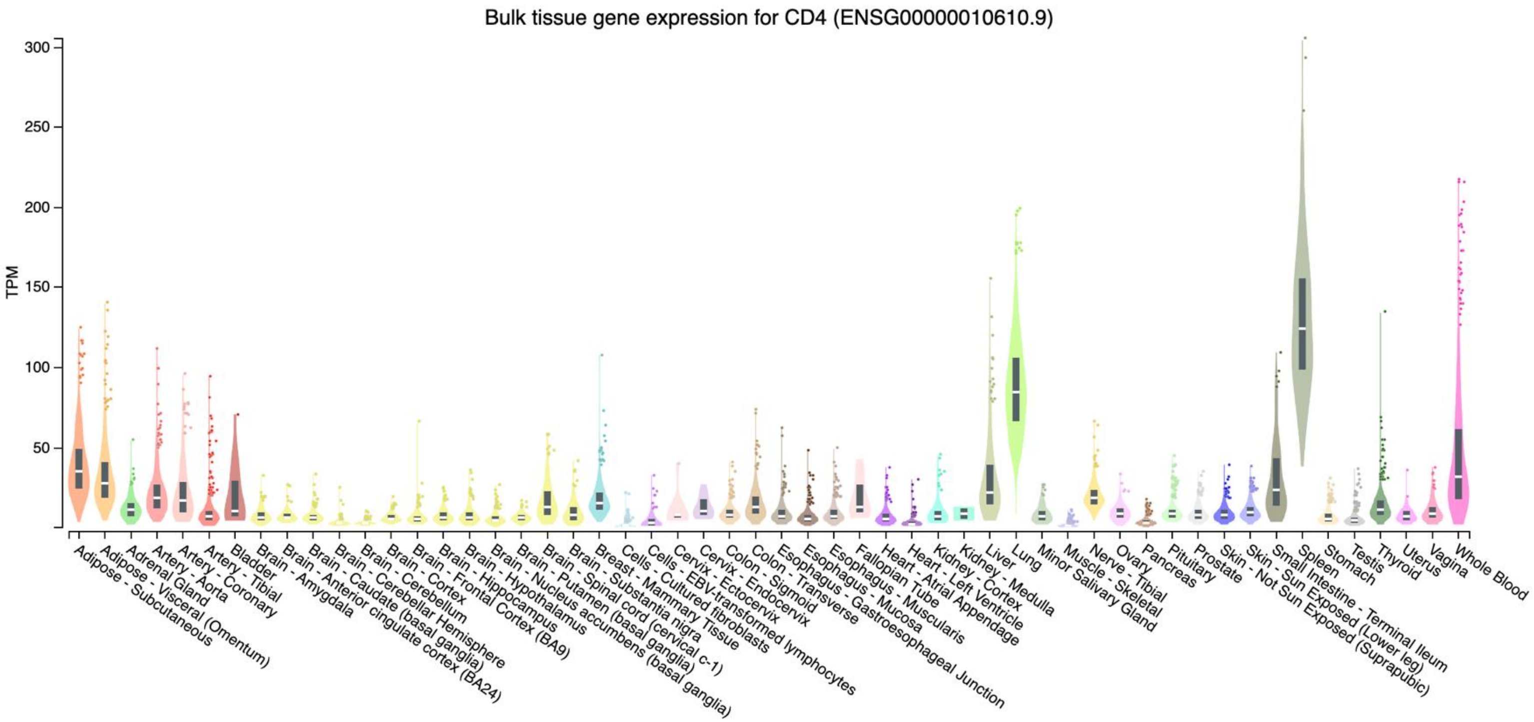


4

RNA-sequencing

Why using RNA-sequencing

- Previously, we wanted to assess DNA variation, which is common to the vast majority of all cells
- All genes *are not expressed the same* in all cells
- Each **cell type** is defined by an **RNA expression profile**
- Some **diseases** are characterised by a change of RNA expression patterns in affected tissue
- RNA measurements are cell-specific



RNA-sequencing

RNA selection

- Removing genomic DNA with DNAse
- We may be primarily interested in mRNA (protein-coding)
- But they only account for 1-5% of RNA pool (rest is ribosomal DNA and ncRNA)

poly-A selection:

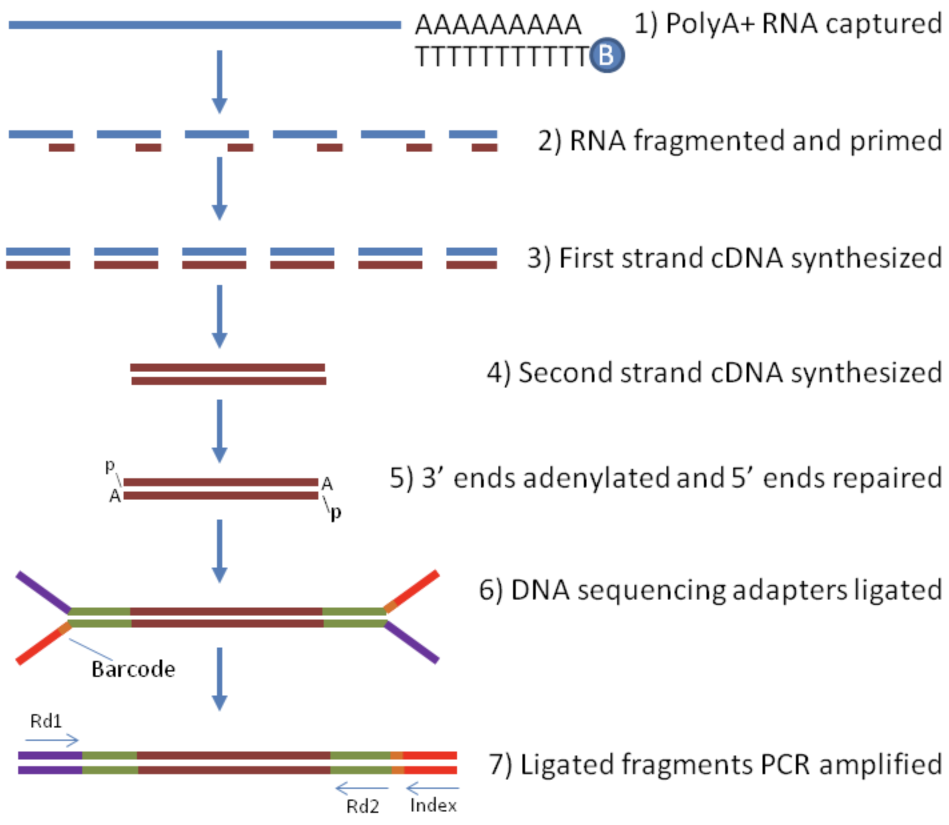
mRNA and most lncRNA contain a poly-A tail

- Selection through hybridisation

Ribosomal RNA depletion:

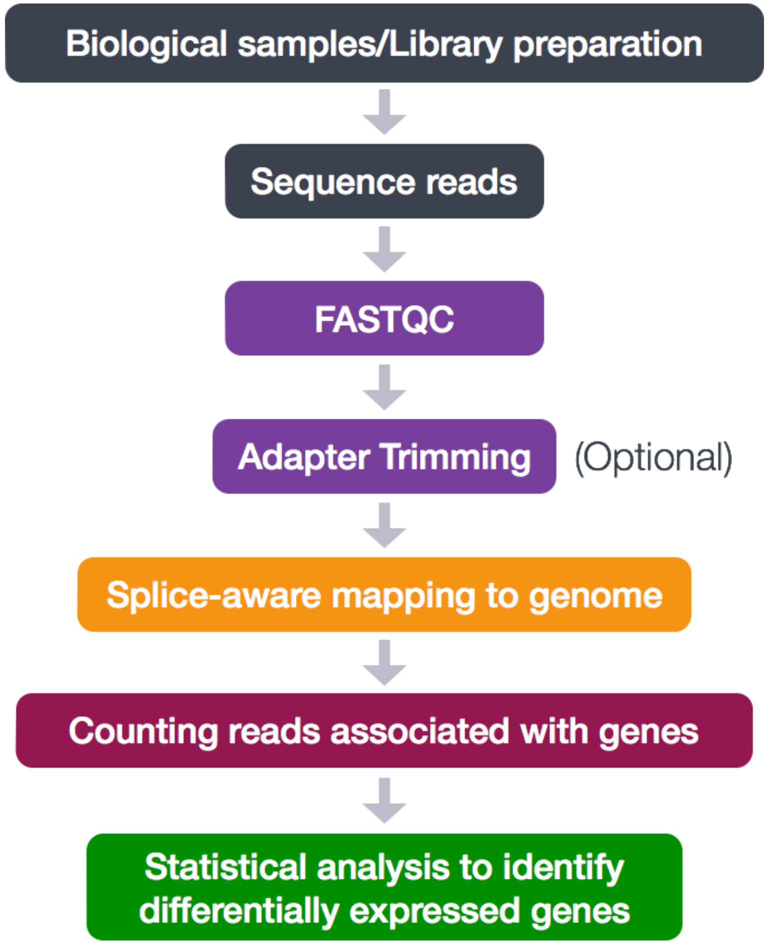
If focus on noncoding RNA or degraded sample:

- Removal of ribosomal RNA by hybridisation to sequence-specific probes



RNA-sequencing pipeline

- Splice-aware alignment – STAR2
- Feature counts - HTSeq
- We are interested in comparing read count per gene isoform between tissues (Differential analysis – DESeq2, Limma, edgeR)
- But many factors influence the total number of reads mapping to a gene
  - GC content
  - Gene/isoform length
  - Experiment-specific variables (total number of reads)
  - …
- Complex read count normalisation is needed



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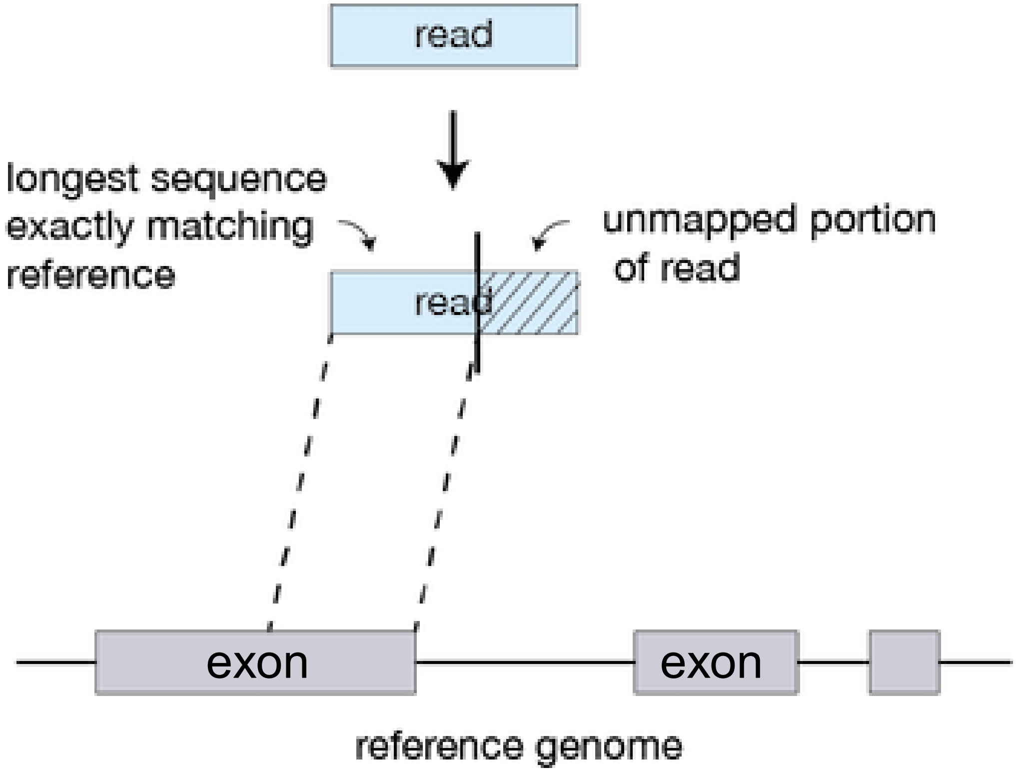
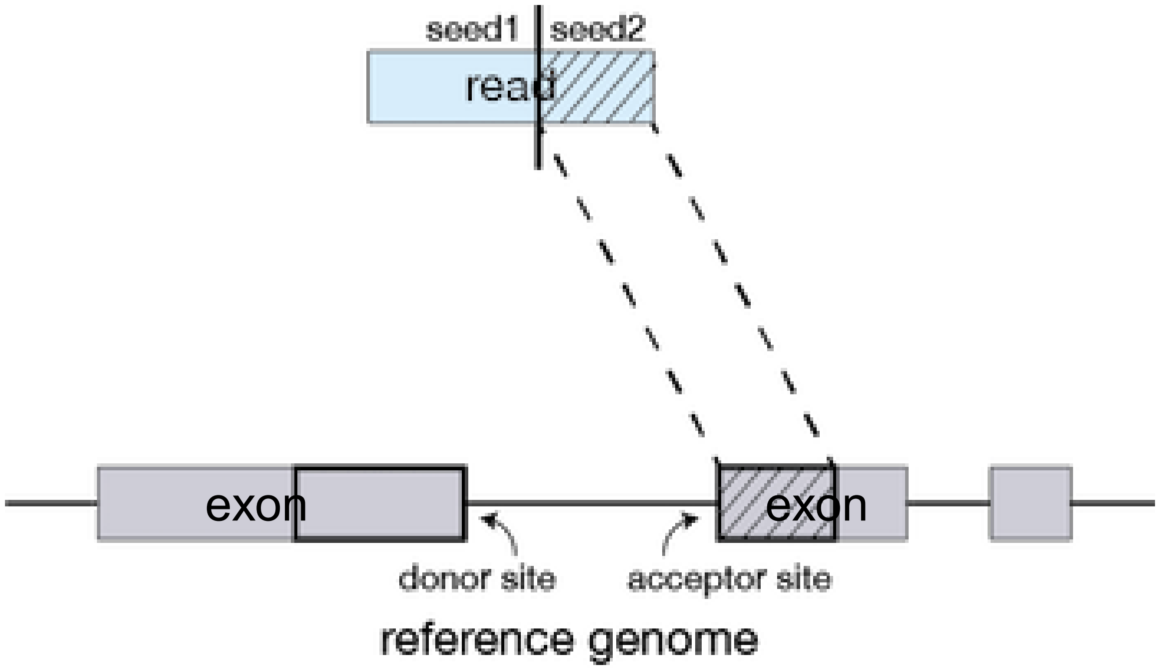
https://hbctraining.github.io/Intro-to-rnaseq-hpc-O2/lessons/03_alignment.html

Splice-aware alignment – STAR2

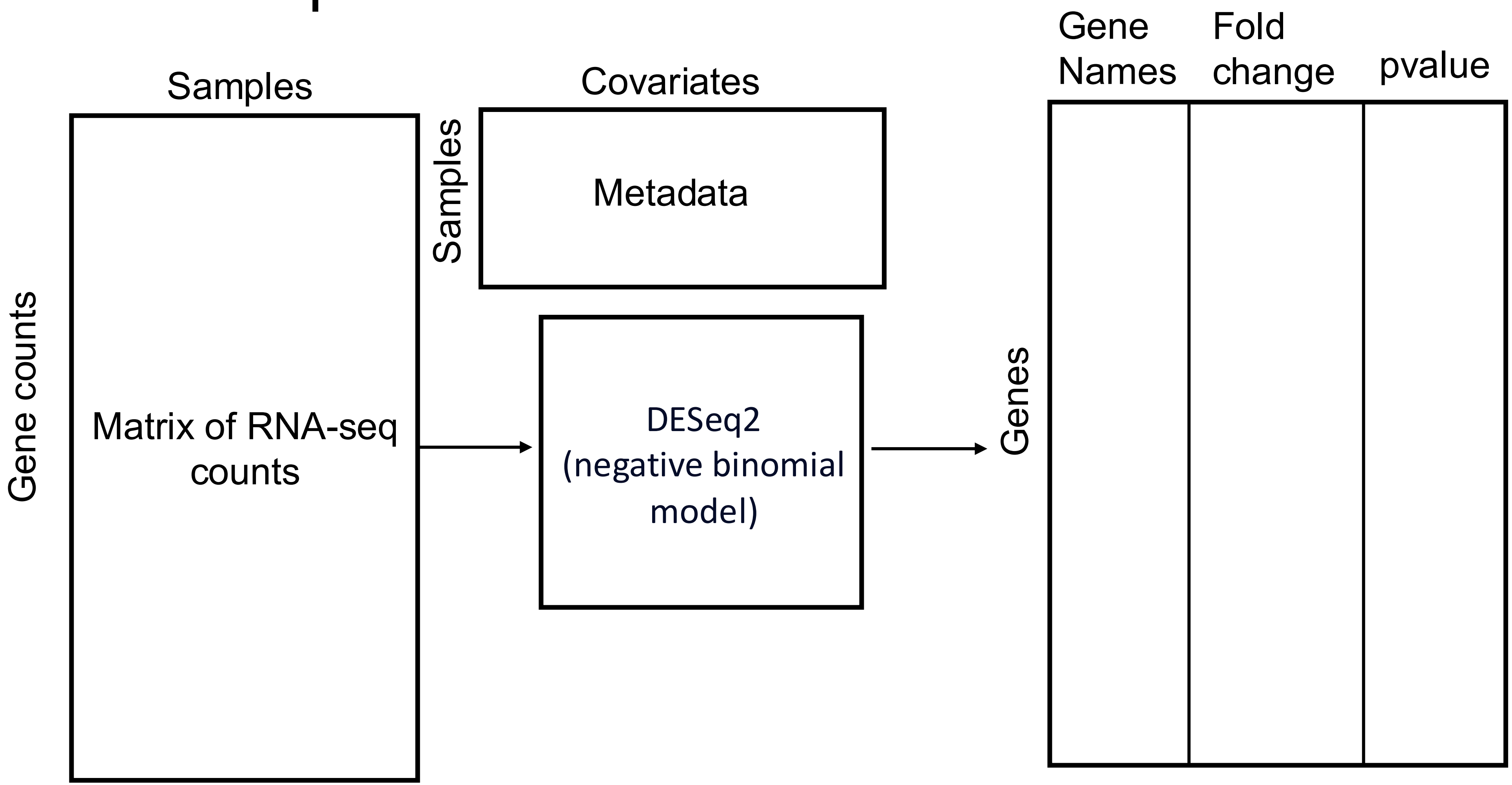




A single read gets split into two alignment 'segments'

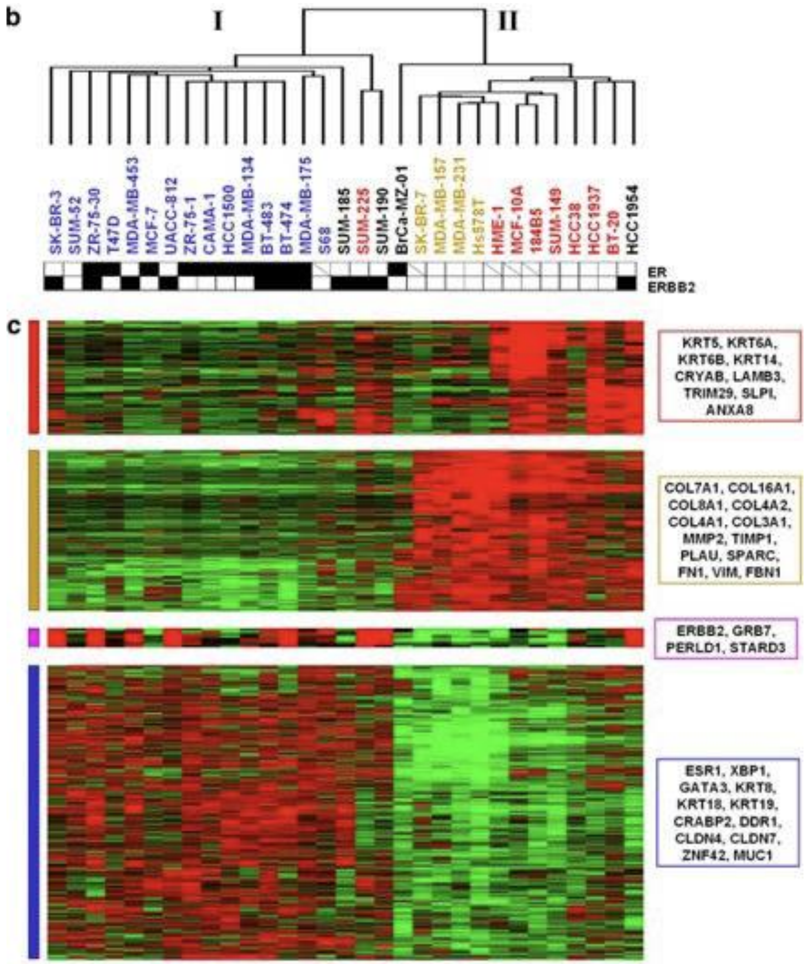
https://hbctraining.github.io/Intro-to-rnaseq-hpc-O2/lessons/03_alignment.html

Differential gene expression (DGE) analysis, typical input and output



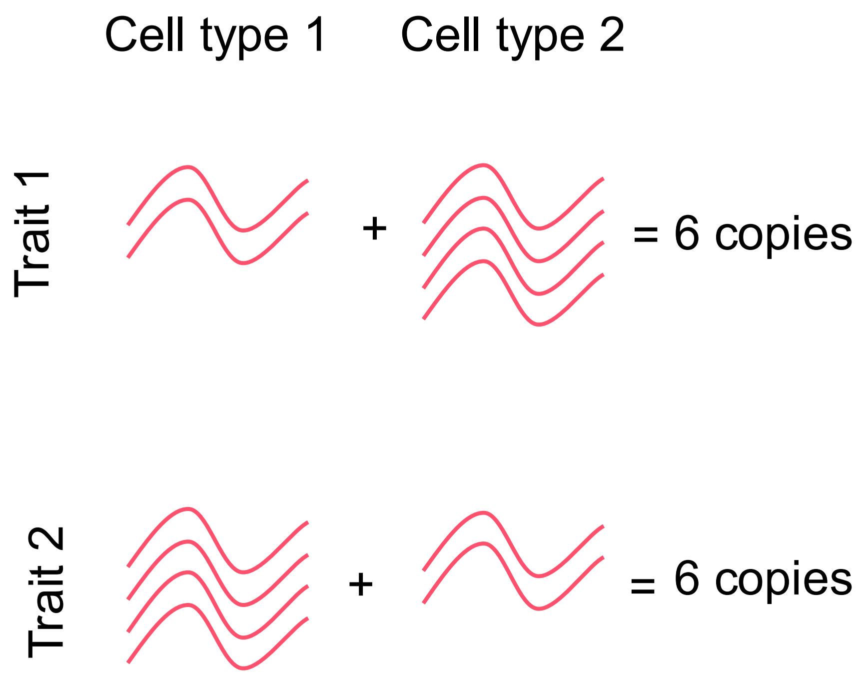
Use genes identified from DGE for clustering

This is a typical use case scenario. The genes identified by DGE analysis are visualised with a heatmap. Hierarchical clustering, or other follow-up analyses, on the gene expression can also be used for separating the groups based on the identified genes.



5. Single-cell sequencing

Why single-cell sequencing?



- Bulk RNA-sequencing is measuring the average number of copies of mixed cell population

Why it matters:

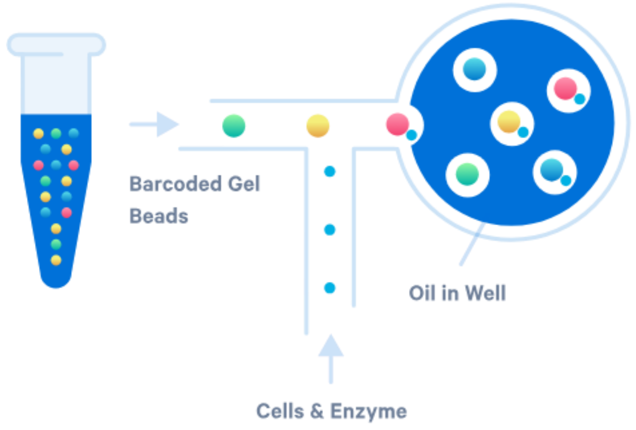
- The disease causing gene might only be altered in a specific cell type
- For DNA sequencing, only a specific cell type might carry the mutation, for example cancer vs healthy tissue

Single-cell (RNA) sequencing: 10X genomics

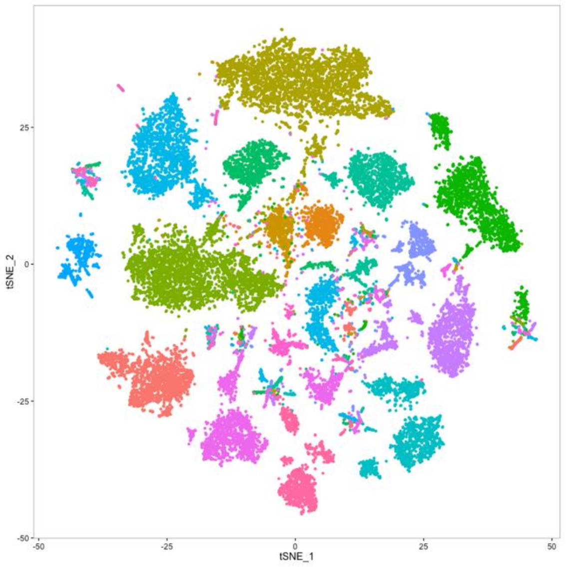
Principle: each cell gets its own tiny reaction chamber.

How it works:

- Cells, **barcoded gel beads**, and oil flow through a microfluidic chip
- Each droplet (**GEM**) ideally contains **one cell + one bead**
- Every RNA from that cell gets tagged with the **bead's barcode**
- All GEMs are pooled and sequenced together - barcode reveals which cell each read came from



Key trick: load few cells → only **1–10% of GEMs** contain a cell. This minimises **doublets** (two cells in one droplet, which look like fake chimeric cell types).



Limitations:

Low gene coverage per cell (~1,000–5,000 of ~20,000 genes detected)

Complex bioinformatic pipeline (clustering, UMAP, batch correction)

Expensive

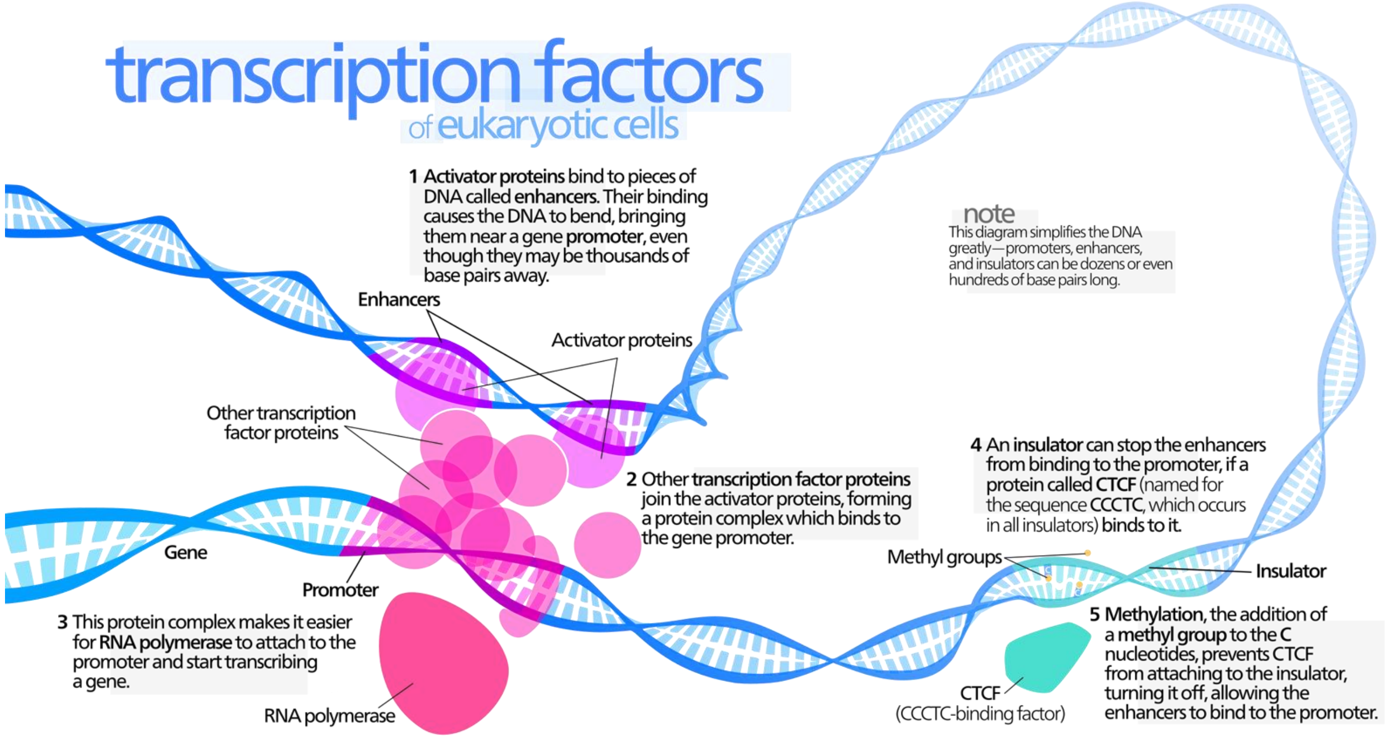
Cell-type annotation needs manual curation

Functional genomics

We looked at how to study DNA variation and gene expression. NGS can be applied to the study of all other functional modifications:

- DNA methylation (bisulfite sequencing, Nanopore sequencing !)
- Histone modification (ChIP-seq, Cut & Run, Cut & Tag)
- Chromatin conformation (Hi-C)
- Open Chromatin (ATAC-seq)
- Protein levels (Proteomics)

The aim is to combine them to understand the context specific modifications that lead to a specific phenotype, for example our disease of interest.



Thank you.